
3D Semantic Splatting: Fusing DINOv3 Features with Gaussian Splatting for Coral Reef Monitoring

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Abstract

Coral reefs face accelerating decline, demanding scalable monitoring tools that can operate with minimal labelled data. This report presents a pipeline that fuses dense semantic embeddings from the DINOv3 self-supervised foundation model with 3D Gaussian Splatting (3DGS) to produce a semantically queryable digital twin of a coral reef without any labelled training data. Two-dimensional validation on the Moorea Labelled Corals dataset confirms that DINOv3 embeddings capture meaningful semantic structure in underwater imagery, with the ViT-7b architecture demonstrating consistently superior discriminability over ConvNeXT-L across all evaluated metrics (two DINOv3 backbone architectures). Projecting these features into 3D via an occlusion-aware, confidence-weighted aggregation pipeline yields a semantically coherent 3D feature field, achieving a mean split-half cosine similarity of 0.899 across 2.8 million Gaussian primitives. Few-shot segmentation in 3D against manually labelled reference points reaches 88% of the human performance ceiling for binary coral versus substrate segmentation and 83% at fine class granularity using only ten reference points per class. Unsupervised clustering produces ecologically interpretable partitions at multiple resolutions, and a single-query semantic search tool enables interactive zero-shot 3D segmentation from a single pixel selection. These results establish DINOv3 and 3DGS as a viable foundation for label-efficient, large-scale coral reef monitoring, though generalisability beyond the single reef site evaluated remains to be validated. The work has been accepted for presentation at the quadrennial International Coral Reef Symposium (ICRS 2026). Interactive demonstrations of the resulting semantic 3D models are available at Semantic-Splatting.

Ethics and Sustainability Statement

This project fits within the scope of ethics pre-approval process, as reviewed by my supervisor Jeff Clark and approved by the faculty ethics committee as application 16028.

Statement on the Responsible Use of AI

In the development of this work, AI (such as ChatGPT) was used as follows:

- **Code Development:** AI was used to assist in debugging and optimising code, e.g., improving figure formatting, ensuring efficient CUDA memory management, etc. All AI-generated code was manually reviewed and tested.
- **Writing and Editing:** AI was used only for proof-reading and grammar corrections. All analysis is original work.

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1 Introduction

Coral reefs are among the most biodiverse and valuable ecosystems on the planet, yet they are acutely vulnerable to the accelerating impacts of climate change. Rising sea temperatures and ocean acidification are driving mass bleaching and mortality events [1]. Although coral reefs cover less than 0.2% of the ocean floor, they support approximately 25% of marine species, making the fish fauna of coral reefs two orders of magnitude more diverse than the surrounding oceans [2]. Beyond their environmental significance, reefs provide vital goods and services such as seafood, coastal protection and tourism which underpin the food security and economies of millions of people globally [3]. Safeguarding coral reefs is therefore essential to preserving both global biodiversity and human livelihoods.

Coral reefs face a multitude of threats, the foremost being climate change. Ocean temperatures are projected to accelerate until net-zero emissions are achieved [4], making this an ongoing and growing concern. Additional pressures such as pollution, destructive fishing practices, coastal development and disease further accelerate reef degradation [5]. Current predictions forecast a majority (> 60%) of individual coral reefs will be depleted to a state where live coral accounts for less than 5% of the total reef substrate by the end of the century if warming trends persist [6]. The rapid and multifaceted decline of coral reefs presents the need for efficient, comprehensive monitoring methods that are capable of capturing changes at appropriate spatial and temporal scales. Such methods are vital to guide effective management and restoration strategies.

Traditional coral monitoring methods, such as photo-transects or quadrats, require expert labour, which limits their practicality on a large scale. In recent years, close-range photogrammetry, which involves the reconstruction of three-dimensional scenes from overlapping two-dimensional imagery, has emerged as a transformative tool in coral reef ecology monitoring. Photogrammetry allows millimetre-scale mapping of reef topography and colony structure over large continuous areas, which enables detailed analysis of coral composition and habitat complexity. With these large detailed datasets, coral restoration projects can be made more effective.

However, the three-dimensional models created by photogrammetry are typically represented as meshes or point clouds. These representations, while geometrically accurate, are poorly suited for semantic analysis. Meshes are topologically complex and difficult to label consistently, while point clouds lack the spatial continuity required for dense feature integration. Neither supports real-time rendering at the scales required for interactive ecological inspection. Consequently, the abundance of high-quality photogrammetric data has shifted the bottleneck from data acquisition to semantically meaningful analysis.

A more suitable representational foundation is provided by 3D Gaussian Splatting (3DGS) [7], a technique that models a scene as a collection of millions of Gaussian primitives, each parametrised by position, orientation, colour and opacity. Unlike meshes or point clouds, 3DGS supports real-time, photorealistic rendering while maintaining an explicit, differentiable scene representation. Critically for this work, the explicit nature of Gaussian primitives makes it straightforward to attach auxiliary per-point attributes (such as semantic feature vectors) directly to the scene geometry. This makes 3DGS a natural foundation for a semantically-rich coral reef digital twin.

A further challenge is the semantic analysis of this data. Neural networks have been used to great effect for the purposes of fast coral segmentation and classification. However, these supervised methods are limited by scarce labelled data, making use in large-scale, cross-regional monitoring programs less feasible. This project aims to address this generalisation gap using self-supervised foundation models, which offer a powerful alternative. Specifically, using the rich dense visual embeddings produced by DINOv3 [8], the latest generation of the DINO family of self-supervised Vision models. The DINO framework, originally introduced by Caron et al. [9] and substantially advanced through DINOv2 [10], has demonstrated a remarkable ability to learn semantically meaningful patch-level representations without any labelled supervision. Crucially, prior work has demonstrated that DINOv2 embeddings are effective for coral image segmentation in sparse-label settings [11], bridging the domain gap of the underwater environment. DINOv3 represents a further advancement of this architecture, yet it has not been evaluated for coral analysis, nor has any DINO-family model been integrated with 3D scene representations for reef monitoring.

This project therefore proposes a novel pipeline that fuses DINOv3 semantic embeddings with a 3DGS

reconstruction to produce a semantically-rich, queryable digital twin of a coral reef. By projecting high-dimensional patch features extracted from survey imagery directly into the Gaussian primitives of an existing 3DGS model, each point in the 3D scene is endowed with a rich semantic descriptor derived without any labelled training data.

The contributions of this project include validation that DINOv3 embeddings capture semantically meaningful structure in underwater coral imagery under sparse-label conditions. Building on this, a 2D-to-3D projection pipeline that lifts these features into a 3DGS representation is implemented. The semantic coherence and discriminative quality of the resulting 3D feature field are then evaluated through split-half consistency and unsupervised clustering. A coarse labelled dataset was also created to quantitatively validate 3D segmentation performance. Finally, an interactive semantic search tool enabling one-shot 3D segmentation from a single pixel query is demonstrated, providing practical proof of concept for a label-efficient reef monitoring strategy. Overall, these contributions represent a step toward semantically-aware coral reef monitoring that scales without the labelling burden of existing approaches.

Section 2 reviews the relevant literature spanning coral monitoring methods, photogrammetric reconstruction, foundation models, and semantic scene representations. Section 3 describes the datasets used for both 2D validation and 3D projection. Section 4 details the full processing pipeline. Section 5 presents the quantitative and qualitative results, and Section 6 contextualises these findings and outlines directions for future work.

2 Literature Review

Coral reef monitoring has traditionally relied on in-situ manual techniques such as belt transects, quadrats, and chain-and-tape surveys, operating across a range of spatial resolutions (Figure 1). While providing valuable data, these methods are labour intensive, costly, and require a high level of taxonomic expertise [12]. Finer-resolution techniques, such as quadrat sampling, that capture the detail needed for structural analysis can only feasibly cover small areas. Additionally, coral grows at very slow rates ($cm \cdot yr^{-1}$) [13], making manual detection of fine-scale structural changes for large areas of reef highly challenging. Common metrics such as linear extension rate offer some insight into growth but fail to account for structural complexity. Alternatively, the chain and tape method can be used to measure surface irregularity (rugosity), but it is potentially damaging to colonies. Consequently, researchers have turned to photogrammetry, which offers a faster, more accurate, and non-invasive option [14].

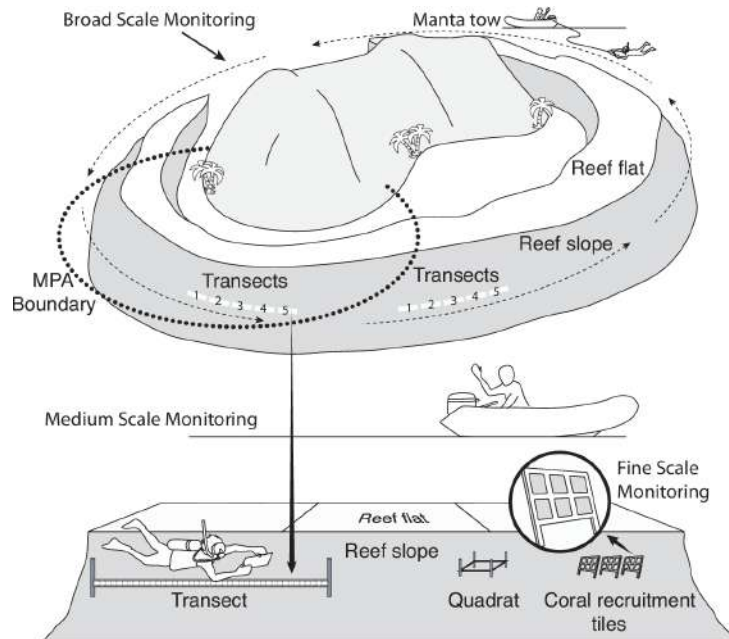


Figure 1: Three scales of coral reef monitoring: broad-scale surveys (e.g. manta tow), medium-scale transect methods, and fine-scale quadrat and recruitment sampling. Reproduced from Hill and Wilkinson [12].

Close-range underwater photogrammetry has been established as an invaluable tool for coral monitoring, offering greater efficiency and ecological insight than traditional methods. Its use has increased as appropriate underwater cameras have become more cost-effective and accessible over the last two decades [14]. A key benefit of photogrammetry is the comprehensive nature of the data provided, being able to capture great detail of ecosystem states and dynamics. In contrast, traditional methods can only capture a small subset of this detail [15]. Photogrammetry, therefore, is able to provide valuable metrics like structural complexity in a fast and non-invasive manner, making it far more practical for large-scale coral monitoring than previous methods.

The efficiency of photogrammetry has led to a shift from a lack of high-quality data to an abundance, creating a new bottleneck in data analysis. A prerequisite to many data analysis techniques is manual labelling, a process which requires trained ecologists to manually segment thousands of images or classify massive numbers of individual points by their species or health state. This type of labelling is infeasible at scale [16]. This issue of insufficient labelled ground truth data is a significant focus of recent research, especially in the context of coral monitoring. Analysis of coral imagery is subject to major challenges due to the underwater setting, including variable imaging conditions, the diversity and coexistence of coral species, lighting variability, and water column effects [17]. These factors are necessary to consider in any deep learning approach for the purposes of coral analysis. Automated coral analysis, resilient to sparse ground truth data, is therefore a problem that requires attention.

Early efforts to automate 2D coral analysis using traditional image classification methods like Support Vector Machine (SVM) demonstrated the viability of automated systems, but struggled to overcome the challenges of the underwater setting [18]. Modern deep learning architectures have since been shown to outperform these methods for segmentation of coral imagery [19], both using patch classification based on random point annotation and semantic segmentation based on pixel-level labels [20]. However, these methods are not able to demonstrate spatio-temporal generalisation and many are still reliant on sizeable labelled datasets, limiting their scalability to only reefs with existing in-situ labelled datasets [21]. As a result, more recently foundational models have been explored as a potential means to reduce the need for such quantity of domain specific data.

The rise of a new class of foundational models, such as the Segment Anything Model (SAM) [22] and Vision Transformers (ViTs) presents this new opportunity in coral analysis. Pre-trained on massive, diverse datasets, they can produce rich semantic embeddings that capture nuanced visual information about unfamiliar objects and scenes. This allows for zero-shot approaches to learning and for analysis of coral reef structure and composition. However, SAM has been shown to perform poorly for coral segmentation of large patches [23]. To solve this, CoralSCOP offers a foundational model for dense coral segmentation which is based upon SAM [24], but trained specifically on a large coral dataset. This results in high performance for coral segmentation and generalisation across locations due to its variety of training data. Though powerful, its application is limited to segmentation tasks. On the other hand, human-in-the-loop segmentation using the ViT DINOv2 foundation model [10] proves effective without any fine-tuning to perform point label propagation in underwater imagery [11]. They demonstrated further, that this approach significantly outperforms other methods in sparse-label settings. This work shows that recent foundational models are able to bridge the domain gap of the underwater setting. However, de-noising of the DINOv2 embeddings, along with a ‘smart labelling regime’ were required for maximum performance. Subsequently, a more powerful DINOv3 model has been developed [8]. Therefore, it is likely DINOv3 can provide strong efficacy for coral analysis.

The embeddings provided by these models are not specified for segmentation, so further work is needed to investigate their performance across analysis tasks, such as identification of bleaching. However, all of the methods discussed so far operate exclusively in 2D, analysing individual images or propagating labels across flat image collections. While this is valuable for classification and coverage estimation, it cannot capture the three-dimensional structural complexity that is central to reef health assessment. Metrics such as rugosity, colony volume, and habitat connectivity are inherently spatial properties that require a 3D representation [13]. Bridging 2D semantic features into 3D therefore presents both an opportunity and a technical challenge, first requiring a robust photogrammetric foundation.

Structure from Motion (SfM) is the foundation of most photogrammetric pipelines, recovering both sparse 3D structure and camera pose simultaneously from unordered image collections. Modern SfM systems such as COLMAP [25] achieve this through sequential feature matching, geometric verification, and bundle adjustment, producing a sparse point cloud and calibrated camera parameters that serve as the

initialisation for downstream dense reconstruction. The quality of this initial reconstruction is critical, as downstream semantic projection pipelines depend entirely on accurate camera intrinsics and extrinsics to correctly map 2D feature observations onto 3D geometry.

Building on SfM, novel-view synthesis methods attempt to reconstruct a fully continuous scene representation from which arbitrary viewpoints can be rendered. Neural Radiance Fields (NeRF) [26] achieved a significant breakthrough by representing scenes as continuous volumetric functions parameterised by a multilayer perceptron, enabling photorealistic novel-view synthesis. However, high computational cost and slow training and rendering speeds limit their practical deployment [27]. More critically for semantic analysis, the implicit encoding of scene content in network weights makes it difficult to attach explicit per-point semantic attributes as there is no discrete geometric primitive to which a feature vector can be assigned. These limitations directly motivated the development of 3D Gaussian Splatting (3DGS) [7], which represents scenes explicitly using millions of Gaussian primitives, each possessing learnable parameters for position, shape, colour and opacity. While this results in large storage sizes for 3DGS, it allows for real-time rendering speeds and fast training times [28]. In addition to this, the explicit representation greatly simplifies the process of mapping feature embeddings onto points in the scene, making it more practical for the purposes of 3D coral analysis. Furthermore, the photorealistic results generated from 3DGS, aid in creating a tool that experts from alternative fields (such as marine ecologists) can easily understand and effectively use. Directly integrating valuable semantic information into these representations would make this of even greater value for interactive analysis.

The challenge of acquiring labelled data, which already limits 2D methods, is even more severe for direct 3D AI analysis [29], hindering the application of 3D-native architectures like PointNet to photogrammetric models. Furthermore, the computational cost is restrictive, often requiring a reduction in precision. As a result, other work has instead successfully demonstrated 2D-to-3D projection of coral segmentation [30]. However, these approaches project discrete class labels from fully trained task-specific segmentation models, foregoing the rich, continuous embedding information that would enable flexible downstream analysis with minimal annotation. This motivates the development of a pipeline that projects dense, high-dimensional feature representations directly into 3D, allowing the full semantic expressiveness of foundation model embeddings to be preserved within the reconstructed scene.

Recent progress combining 3DGS with foundation model features has validated this general direction. GauSSmart uses DINOv3 embeddings as a semantic loss term during 3DGS training to improve geometric and photometric reconstruction quality [31], demonstrating that DINOv3 features carry spatially meaningful signals compatible with the 3DGS training objective. In contrast, Feature-3DGS creates an explicit 3D feature field model by distilling 2D features from models like SAM into a compressed, ‘student’ representation [32]. While powerful, this distillation process introduces non-linear approximation error and sacrifices embedding depth in favour of rendering performance, reducing the richness of the resulting semantic field. Neither approach has been applied to coral reef monitoring, and neither addresses the specific challenge of sparse-label ecological analysis where the full discriminative depth of the feature space is scientifically critical.

This project takes a complementary but distinct approach. Rather than distilling features into a learned compressed representation, PCA compression calibrated explicitly to retain maximum semantic variance is applied, storing a 64-dimensional embedding per Gaussian primitive. This dimensionality is determined empirically by analysing the explained variance of the DINOv3 feature space across reef imagery, retaining the principal components that capture both dominant structural variance and the fine-grained local features necessary for robust downstream segmentation. The result prioritises scientific interpretability and analytical flexibility over real-time rendering performance, directly addressing the critical gap in 3D coral analysis under sparse label conditions.

3 Data

The first dataset used is the Moorea Labelled Corals dataset [33], which contains almost 400,000 human expert random point annotations across 2,055 images captured at the island of Moorea, French Polynesia, between 2008 and 2010 (Figure 2). Labels cover the nine most abundant coral classes alongside four non-coral categories including sand, rubble, and anthropogenic substrate. This dataset was used to validate the semantic quality of DINOv3 embeddings for coral analysis, providing ground-truth labels against

which clustering and few-shot classification performance could be quantitatively assessed.

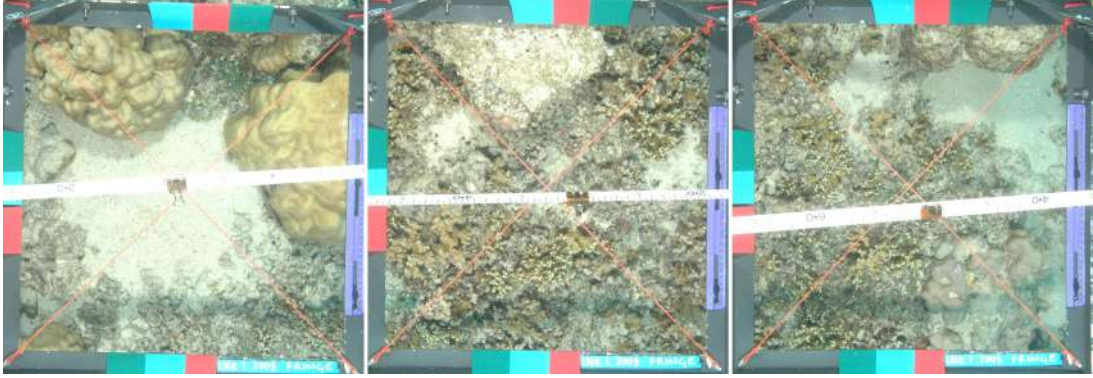


Figure 2: Examples of quadrat images from the Moorea Labelled Corals dataset (1944×1948). Labelled classes are provided in a separate file with pixel coordinates.

The second dataset is the Wildflow sweet-corals dataset [34], specifically the Tabuhan P1 section, which provides a complete photogrammetric pipeline from raw imagery through colour correction (Figure 3a), sparse point cloud reconstruction, and final 3DGS model. Camera poses were estimated using COLMAP [25], with Gaussian primitives optimised using the original 3DGS training procedure [7]. The model was provided pre-trained by Wildflow as a set of spatially chunked sub-blocks, removing the need to reproduce the photogrammetric reconstruction and allowing focus to remain on the semantic projection pipeline.

Each Gaussian primitive is parametrised by position, covariance, opacity, and spherical harmonic coefficients up to degree three. To manage computational requirements, a 5×5 m sub-block was selected for semantic projection, chosen to capture a representative diversity of reef substrate types including coral colonies, rubble, and sand. This sub-block comprises approximately 2.8 million elliptical Gaussian primitives (e.g. Figure 3b) and is covered by 555 source images.



Figure 3: The Wildflow Tabuhan P1 dataset: (a) shows a raw high-resolution source image. (b) visualises a section of the resulting 3D Gaussian Splatting model with their elliptical structure.

4 Methodology

As shown in Figure 4 the methodology is split into two core sections resulting in a dual mode output. The first section establishes a 2D processing pipeline ensuring a consistent and rigorous methodology for processing coral imagery. This allows for validation of 2D segmentation capability to ensure that DINOv3 is able to bridge the domain gap. The second section describes the subsequent 3D processing. The established 2D feature extraction pipeline is applied to source images of an existing 3DGS model,

projecting the resulting feature maps into a 3D semantic representation. Methodology for validation and analysis of both the 2D and 3D outputs of the full pipeline are also described throughout.

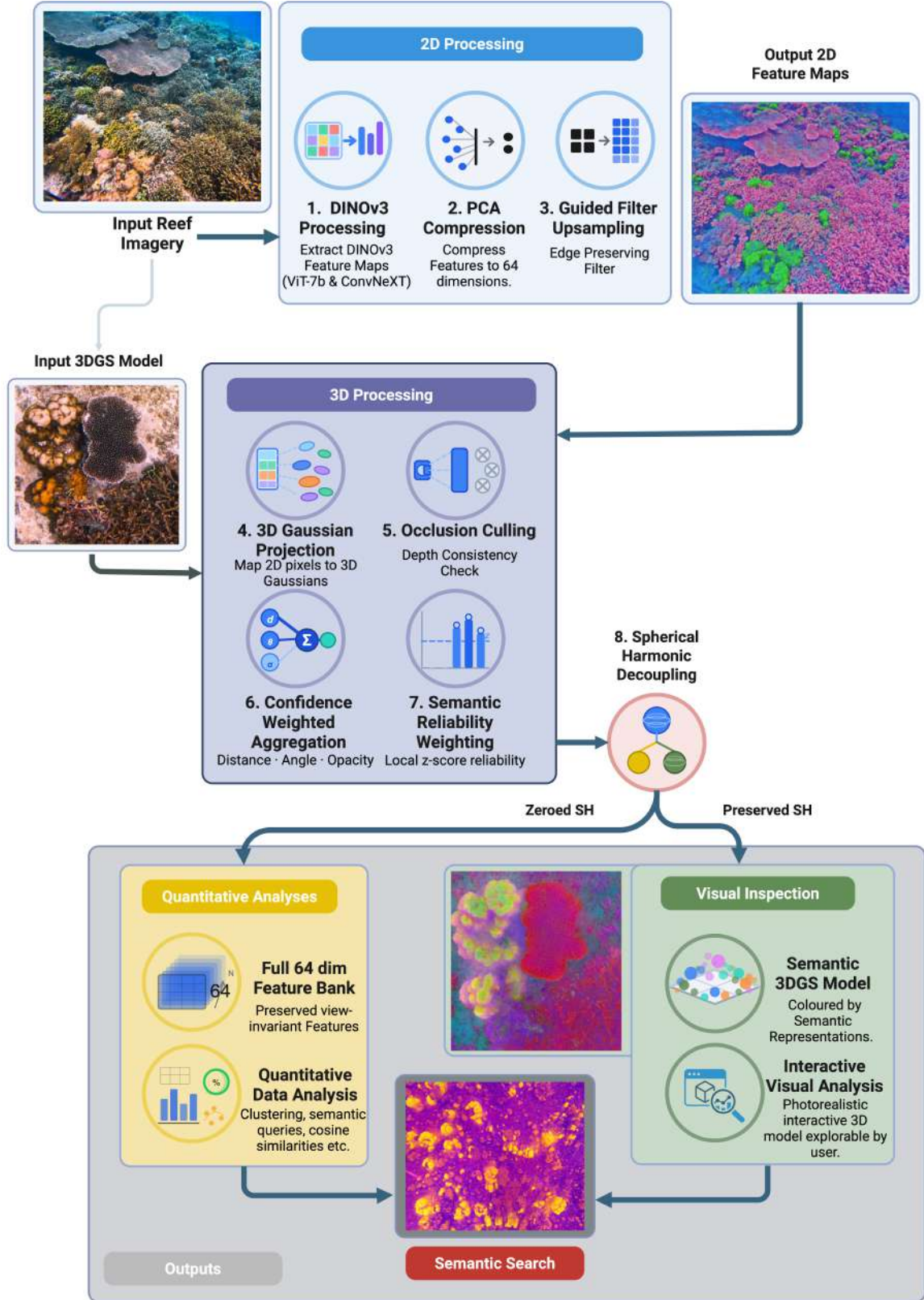


Figure 4: Methodology Overview: demonstrating the two sections of the semantic splatting pipeline (2D & 3D processing) and resulting outputs. Each step is described in detail throughout Section 4.

4.1 Model Selection

DINOv3 was selected as it achieves state-of-the-art performance among self-supervised vision foundation models on dense prediction benchmarks without task-specific fine-tuning [8], making it the strongest available backbone for zero-shot patch-level feature extraction.

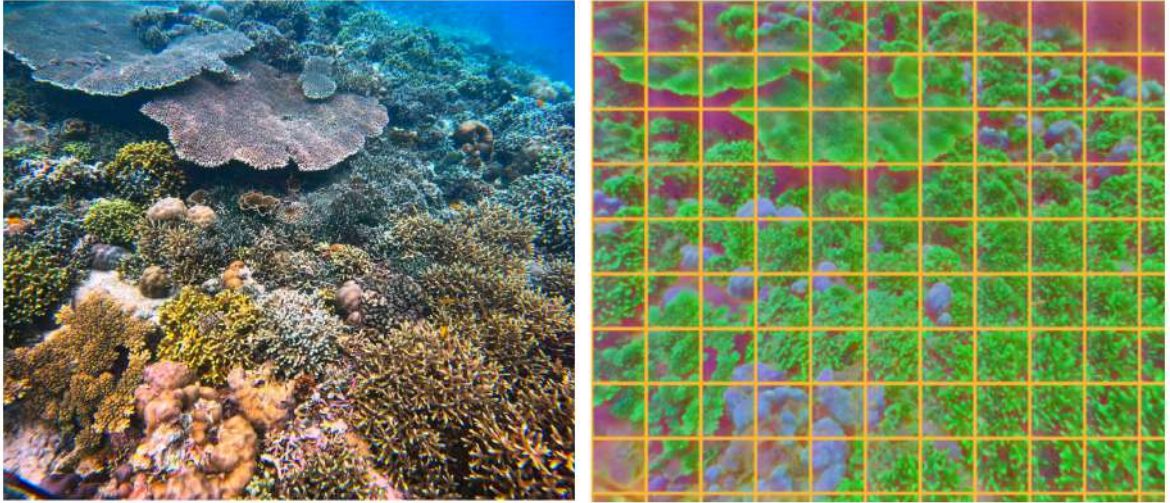
Raine et al. [11] demonstrate the viability of the DINOv2 ViT for coral segmentation; however, the use of DINOv3 and particularly its ConvNeXT architecture which offers a convolutional alternative to the ViT backbone remains unexplored in this context. Therefore, both architectures were evaluated to characterise their respective strengths and weaknesses. The largest available DINOv3 models, the ViT-7b and ConvNeXT-L, were selected to maximise feature quality within each architectural family.

4.2 2D Processing

The 2D processing stage serves two purposes. First, it validates that DINOv3 embeddings capture semantically meaningful structure in underwater coral imagery before committing to the more computationally expensive 3D projection pipeline. Second, it establishes the feature extraction methodology that is subsequently applied to the source imagery of the 3DGS model. This section describes the feature map generation process and the clustering approaches used for 2D semantic validation.

4.2.1 Feature Map Generation

DINOv3 extracts features by dividing the input image into a grid of fixed-size patches and computing a feature embedding for each. For large (5372×4696) images, processing in a single pass is infeasible due to computational constraints, necessitating a sequential patch processing strategy. This approach aligns with the translational equivariance of the ConvNeXT model’s architecture, producing a coherent feature map. In contrast, the ViT’s self-attention mechanism is global, meaning pixels at the patch border are context-starved as they cannot see pixels beyond the edge. This produces noise in the feature calculation at the border of each patch, which, when combined, results in severe seam artifacts seen in Figure 5b.



(a) Raw Source Image (5372x4696).

(b) ViT-7b Embedding PCA (335x293)

Figure 5: Demonstration of severe seam artifacts (orange grid in b) resulting from a non-overlapping processing strategy for the ViT model.

To solve this, a sliding window (256×256) with a 50% overlap ($stride = 128$) was used. In the overlapping regions, a pyramid weight mask was applied, which gives a high weight to central features, and low weight to edge features. The mask is defined over normalised coordinates (x, y) where $(-1, -1)$ is a corner and $(0, 0)$ is the centre, as follows:

$$W(x, y) = (1 - |x|)(1 - |y|). \quad (1)$$

The final feature vector for a given pixel coordinate p is the weighted average of all overlapping patch features at that coordinate:

$$F_{\text{final}}(p) = \frac{\sum_i W_i(p) \cdot F_i(p)}{\sum_i W_i(p)}. \quad (2)$$

where $i \in \mathcal{P}$, and $\mathcal{P}$ is the set of all patches representing all overlapping sliding windows.

The implementation of this pyramid-weighted blending strategy ensures a theoretically continuous feature field by prioritising high-context central pixels and interpolating latent representations across overlapping boundaries. While this approach effectively mitigates the context starvation inherent to ViT patch-processing, it introduces a substantial computational bottleneck. Empirical testing on a 16GB VRAM GPU resulted in approximately $60\times$ longer processing times compared to the ConvNeXT model.

4.2.2 Clustering Methodology

The segmentation capabilities of the models were evaluated on the Moorea Labelled Corals dataset, comparing a simple unsupervised k-means approach against a supervised KNN approach. The supervised test allows for evaluation of whether the features align well with ground-truth labels, while unsupervised tests how well the features capture meaningful semantic clusters without external guidance.

Unlike the 3D projection pipeline where PCA compression is applied to manage memory overhead (Section 4.3.2), the 2D clustering experiments operate directly on the full-dimensional feature embeddings which are 4096 dimensions for the ViT-7b and 1536 for the ConvNeXT-L. This ensures that the 2D validation results reflect the maximum discriminative capacity of the models without compression artefacts.

A critical pre-processing step was the application of feature-centring to account for the high anisotropy typically found in self-supervised embedding spaces [35]. Because these features often occupy a narrow directional cone, raw cosine similarities can be artificially inflated, reducing the discriminative resolution between classes. To mitigate this, a feature centring and normalisation strategy similar to the stability mechanisms described in the original DINO framework is applied [9]. The global mean μ_F of the feature space was subtracted from each vector, this re-orientes the origin of the coordinate system to the centre of the data distribution, allowing subsequent L2-normalisation to effectively span the full hypersphere. Following centring, all feature vectors were L2-normalised such that

$$\hat{v} = \frac{v - \mu_F}{\|v - \mu_F\|}. \quad (3)$$

This allows the cosine similarity to be calculated using a single matrix multiplication where

$$S(\hat{A}, \hat{B}) = \hat{A} \cdot \hat{B}. \quad (4)$$

By centring before normalisation, the clustering algorithms operate on a zero-mean hypersphere, ensuring Euclidean distances are directly proportional to cosine distances.

For the unsupervised k-means approach, the algorithm was initialised to find $k = 6$ clusters, chosen to match the number of classes in the ground-truth labelled dataset for the images tested, thus allowing for a more direct qualitative comparison between the supervised and unsupervised results.

To improve the robustness of the supervised KNN clustering, two constraints were applied. To filter labelled point outliers, for each class the centroid (C_j) of each given group of labelled feature vectors was calculated. If the cosine distance from the centroid of any vector (v_i) is greater than half of a standard deviation from the mean distance,

$$1 - S(v_i, C_j) > (\mu_D + 0.5 \cdot \sigma_D). \quad (5)$$

this labelled point is excluded. This mitigates errors in the labelled dataset, for example Figure 10a, shows several points on the white tape measure that were incorrectly labelled as sand in the dataset. These points varied significantly from the semantic centroid of all sand labelled points, so were correctly excluded as outliers. However, some correctly labelled points were excluded as outliers, such as the pink points in the bottom right of the image, due to natural intra-class appearance variations. This issue would be further mitigated in practice by a more rigorous labelling protocol, or greater fine tuning of the

exclusion threshold. The second constraint applied was if any pixel’s greatest similarity to a class was below a conservatively chosen threshold ($\tau = 0.2$),

$$S_{\max}(v_p) < \tau, \quad (6)$$

then it was assigned to the ‘unknown’ class, preventing points unique to any labelled class from forceful classification. This allows points that are genuinely unique to any labelled class to be separately classified.

4.2.3 Few-Shot Classification

To evaluate the models’ utility in a realistically sparse labelling scenario, a few-shot classification test was conducted on the Moorea Labelled Corals dataset. Rather than providing per-image reference points, a small number of globally sampled labelled points were selected per class across the full dataset, reflecting the practical scenario where a taxonomist annotates a limited number of examples once and expects the model to generalise across all survey imagery.

For each class, k reference points were sampled globally, where $k \in \{5, 10, 25\}$ to match the sparse label counts used by Raine et al. [11]. The feature vectors corresponding to these reference points were extracted and used to construct a per-class semantic prototype by computing the mean feature vector across all k examples. All remaining labelled points in the dataset were then classified by assigning each to the class whose prototype exhibited the highest cosine similarity to the query feature vector, subject to the unknown-class threshold defined in Eq. 6. Classification performance was evaluated using the weighted F1-score across all classes, with weights proportional to class frequency to account for the natural class imbalance present in the dataset. This evaluation was repeated for both the ViT-7b and ConvNeXT-L architectures to characterise how few-shot performance scales with label count and varies across model architectures.

Reference points were selected uniformly at random from the available labelled points for each class. A diversity-maximising or spatially stratified selection strategy would likely yield more representative prototypes and more stable results. However, it reflects a realistic baseline assumption and provides a conservative lower bound on few-shot performance.

4.3 3D Projection Pipeline

Integrating DINOv3 semantic embeddings into a 3DGS model first requires extracting feature maps from the 2D source images used for the 3D reconstruction, this was repeated for both the ViT and ConvNeXT models. For this study, an existing 3DGS model of the Tabuhan dataset (provided by Wildflow) was used as the base geometry. To manage computational constraints during development, testing was restricted to a 5×5 m sub-block of the larger reef model. This area is mapped by a collection of 555 unique images, where images with fewer than 1% of points contained within the bounds of the sub-block are ignored.

The core projection logic involves mapping each 3D Gaussian in this sub-block back to the corresponding coordinates in the 2D source images. By sampling the DINOv3 embedding vector at these projected pixel locations across multiple views, a consolidated 3D feature vector can be computed for every Gaussian. In its simplest form, this aggregation is an average of all visible 2D features; however, as detailed in subsequent sections, accurate projection requires accounting for occlusion, distance and image resolution. Where in the context of 2D-to-3D projection occlusion refers to the visual obstruction of background geometry by foreground objects along the camera’s line of sight. The mapping and aggregation of features is repeated for every Gaussian within the section of reef model.

4.3.1 Projection Logic

The projection of a 3D Gaussian g_i with the world-space position $P_i = [x, y, z, 1]^T$ onto the 2D image plane of camera j is governed by the pinhole camera model [36]. This involves a two-step transformation, with an extrinsic transformation followed by an intrinsic projection.

The point P_i is first transformed into the camera’s local coordinate system (Camera Space) using the

extrinsic matrix $E_j \in \mathbb{R}^{4 \times 4}$, which represents the camera’s orientation and position such that,

$$P_{cam} = E_j P_i = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} P_i. \quad (7)$$

Where $R \in \mathbb{R}^{3 \times 3}$ is the rotation matrix and $t \in \mathbb{R}^3$ is the translation vector. The resulting vector $P_{cam} = [x_{cam}, y_{cam}, z_{cam}, 1]^\top$ defines the Gaussian’s position relative to the camera centre, where z_{cam} represents the orthogonal depth from the image plane.

To map the 3D camera-space coordinates to 2D pixel coordinates $u = [u, v]^\top$, the camera intrinsic matrix K is applied such that,

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K \begin{bmatrix} I & 0 \end{bmatrix} P_{cam}, \quad (8)$$

where $s = z_{cam}$ is a scalar factor arising from the homogeneous representation. The matrix K contains the focal lengths (f_x, f_y) and the principal point (c_x, c_y) . Expanding this, the pixel coordinates are calculated using the perspective divide,

$$u = \frac{f_x \cdot x_{cam}}{z_{cam}} + c_x, \quad v = \frac{f_y \cdot y_{cam}}{z_{cam}} + c_y. \quad (9)$$

Because the DINOv3 feature maps F_j are stored as tensors, the pixel coordinates (u, v) must be normalised to a range of $[-1, 1]$ to facilitate differentiable sampling via bilinear interpolation. This transformation ensures that the 3D Gaussian can accurately sample semantic information even when its projection falls between discrete feature patches.

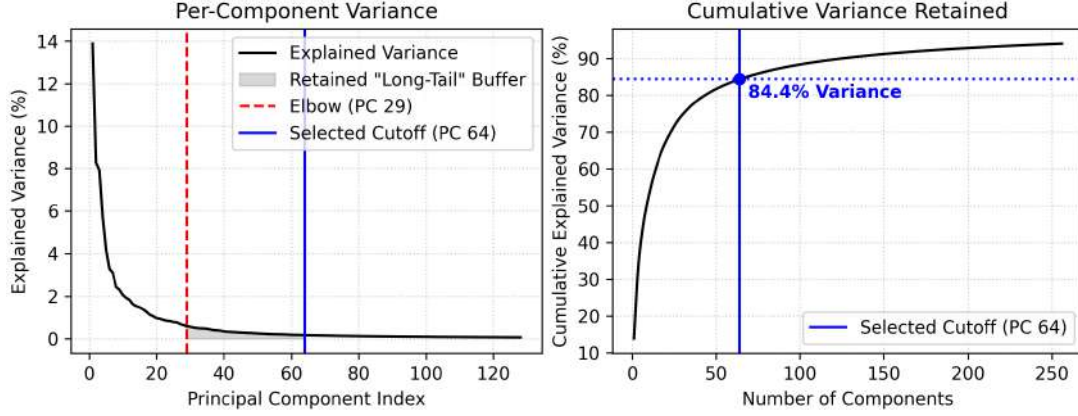
Any point where $z_{cam} \leq 0$ (behind the camera) or where (u, v) falls outside the image dimensions is immediately discarded from the aggregation process. This geometric foundation ensures that the semantic vectors F_{ij} used in the weighted average (Section 4.3.4) are spatially accurate and contextually relevant.

4.3.2 Principal Component Analysis (PCA)

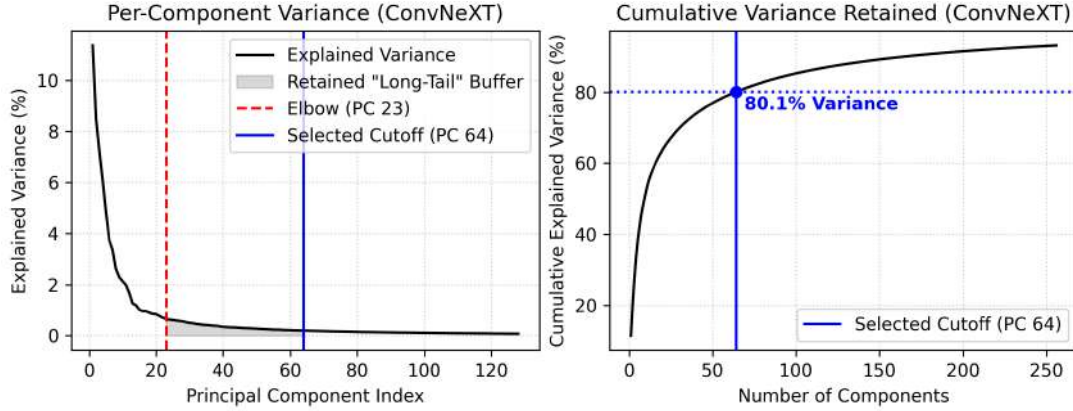
A significant challenge in lifting DINOv3 features to a 3DGS representation is the memory overhead associated with high-dimensional features. A single image’s embeddings with the native ViT-7b dimensionality ($D=4096$) requires approximately 700 MB of storage. Scaling this to the full selected area of 555 images would necessitate over 400 GB of storage. This far exceeds available hardware capacity within the scope of this project.

To mitigate this, PCA was used to compress the embeddings into a lower-dimensional space. To ensure the PCA projection captures the global semantic variance of the reef environment, stochastic sampling was used. Specifically, 10,000 random pixel features were sampled from a subset of 20 images. These samples were concatenated into a single feature matrix $X \in \mathbb{R}^{(200,000 \times 4096)}$. A separate PCA projection matrix was fitted independently for each architecture, as the ViT-7b and ConvNeXT-L produce embeddings of different dimensionality (4096 and 1536 respectively) and different statistical structure. In both cases, the projection matrix was fitted on stochastically sampled features from the Tabuhan dataset specifically, ensuring that the principal components capture the semantic variance of the target reef environment.

The level of compression was determined by analysing the explained variance across the principal components. As shown in Figure 6a, applying the Kneedle algorithm to the scree plot identified an elbow (the point of maximum curvature) at the 29th principal component. This indicates that the primary structural variance is captured within the first 29 dimensions, from this a conservative cut-off of $D = 64$ was selected. Retaining 64 components provides a safety buffer that preserves the ‘long-tail’ of the fine-grained local features necessary for robust downstream segmentation, while representing a $\sim 98\%$ reduction in storage requirements. Furthermore, $D=64$ inherently aligns with hardware-level, power-of-two memory optimisations.



(a) ViT-7b PCA Explained Variance Analysis



(b) ConvNeXT PCA Explained Variance Analysis

Figure 6: Comparison of Scree Plots and Cumulative Explained Variance for both ViT-7b and ConvNeXT models. A later elbow point and higher explained variance at cut-off is observed for the ViT-7b model.

The resulting projection matrix $W \in \mathbb{R}^{4096 \times 64}$ was used to transform the full feature maps. Following projection, the transformed tensors were downcast to half-precision (float16), approximately halving the remaining storage cost. Micikevicius et al. [37] demonstrate that FP16 storage preserves model accuracy across a wide range of architectures and tasks without requiring modifications to hyperparameters. This suggests that the reduction in numerical precision introduced here is unlikely to meaningfully impact downstream performance.

To maintain consistency across comparative evaluations, a standardised cut-off of $D=64$ was selected for both architectures. For the ConvNeXT model, 64 dimensions represent an even more conservative preservation of the original feature space due to its lower initial dimensionality of 1536 evidenced by the elbow at PC 23, shown in Figure 6b.

4.3.3 Upsampling

Due to the patch-based architecture of DINOv3, the extracted feature maps are spatially downsampled relative to the original image resolution. For both the ViT and ConvNeXT models each embedding corresponds to a patch of the image rather than a single pixel. In order to correctly project features onto the 3D Gaussian primitives, the feature map must be rescaled back to the resolution of the source image.

A naive solution is to apply bilinear interpolation to the feature tensor. However, this approach can blur feature boundaries because neighbouring patches belonging to different semantic surfaces are averaged together. This is particularly problematic in reef environments where fine structural transitions occur between coral colonies, rock and sand.

To mitigate this, a guided filtering step introduced by He, Sun, and Tang [38] was applied post-

interpolation. The high-resolution RGB image acts as a guidance signal I , allowing the filtering operation to preserve sharp edges aligned with the intensity gradients in the image. The filter assumes a locally linear relationship between I and the output q within a window ω_k defined by

$$q_i = a_k I_i + b_k, \quad \forall i \in \omega_k, \quad (10)$$

where the coefficients (a_k, b_k) are determined by minimizing the squared difference between q and p . Because a pixel i is contained in multiple overlapping windows ω_k , the final output is computed by averaging all possible values of a_k and b_k that cover pixel i such that

$$q_i = \bar{a}_i I_i + \bar{b}_i, \quad (11)$$

where $\bar{a}_i$ and $\bar{b}_i$ are the mean coefficients calculated via a box filter. This process preserves high-frequency spatial structure from the RGB image, while smoothing noise within homogeneous regions of the feature map. The resulting refined features provide a higher fidelity semantic signal.

4.3.4 Weighted Averaging

Since a reconstructed scene is observed from many viewpoints, each Gaussian g_i may be visible in multiple images. Consequently, the semantic representation of a Gaussian must aggregate features from all valid views.

Rather than computing a simple average, a confidence-weighted aggregation was used to prioritise more reliable observations. For each observation of Gaussian g_i in image j , a confidence weight w_{ij} was computed as the product of three geometric factors. Since w_{ij} appears exclusively within the normalised weighted average of Eq. 15, its absolute scale is irrelevant and only the relative magnitudes across views determine the aggregated feature. The weights therefore function as unnormalised importance scores.

The first being distance weighting w_{dist} . Closer observations provide more reliable projections due to reduced reprojection error, following the principle that camera-to-surface distance is a legitimate reliability criterion in multi-view aggregation [39]. Therefore, distance weight is defined as

$$w_{dist} = \frac{1}{\|\mathbf{x}_{cam}\|^2 + \epsilon}, \quad (12)$$

where x_{cam} is the Gaussian position in camera space. Where $w_{dist} > 0$ and decreases monotonically with camera distance.

Secondly, viewing angle weighting w_{angle} is used because observations where viewing direction aligns with the surface normal provide more reliable appearance information. This principle is well established in image-based rendering, where angular deviation between a source ray and the desired viewing direction is the primary criterion for weighting the contribution of input cameras [39]. This is computed using the dot product between the viewing direction $\mathbf{v}$ and the surface normal $\mathbf{n}$ such that,

$$w_{angle} = |\mathbf{v} \cdot \mathbf{n}|. \quad (13)$$

Because both vectors are unit length, $w_{angle} \in [0, 1]$. With a maximum of 1 for frontal observations and zero for grazing angles.

The third factor is opacity weighting w_{opac} . The opacity of each Gaussian primitive is optimised during 3DGS training to represent surface solidity [7], so Gaussians with higher opacity correspond to more stable, well-reconstructed geometry and are expected to carry more reliable semantic observations. The post-sigmoid opacity value is used directly, giving $w_{opac} \in (0, 1)$ by the properties of the sigmoid function.

The final confidence weight is then

$$w_{ij} = w_{dist} \cdot w_{angle} \cdot w_{opac}. \quad (14)$$

Letting F_{ij} denote the feature vector sampled for Gaussian g_i from image j , the weighted mean feature

vector is

$$F_i = \frac{\sum_j w_{ij} F_{ij}}{\sum_j w_{ij}}. \quad (15)$$

4.3.5 Occlusion

In complex environments such as coral reefs, foreground structures frequently occlude background surfaces. Without accounting for this effect, semantic features from occluding objects could be incorrectly projected onto background geometry, degrading the semantic representation.

To prevent this, an occlusion-aware projection step was implemented using depth consistency checks in screen space. Following the projection of a 3D Gaussian into pixel coordinates $((u, v))$ as defined in Section 4.3.1), its orthogonal depth in camera space, z_{cam} , is compared against a discrete depth map constructed from the projected image grid.

Let $D(u, v)$ represent the minimum recorded depth at pixel (u, v) across all valid projected Gaussians. A Gaussian g_i is considered visible from the current camera view if and only if its depth satisfies the condition,

$$z_{cam,i} \leq D(u, v) + \delta. \quad (16)$$

Where δ is a predefined depth tolerance threshold (0.1) chosen empirically to account for minor reconstruction noise and the volumetric nature of the Gaussians but sensitivity to this parameter was not systematically evaluated. If a Gaussian’s depth exceeds this threshold, it is classified as occluded and is excluded from the feature sampling phase. This helps to ensure that only the visible surface geometry inherits the DINOv3 semantic embeddings.

4.3.6 Uncertainty-Aware Semantic Reliability Weighting

Even with occlusion-aware projection, raw semantic feature aggregation in 3DGS can suffer from inconsistencies caused by projection noise, reconstruction inaccuracies, and partial occlusions. These errors may introduce conflicting semantic observations for the same Gaussian when features are aggregated across multiple views. Rather than removing such points entirely, which risks eliminating valid geometry, a soft reliability weighting strategy was adopted.

First, a semantic uncertainty measure was estimated for each Gaussian based on the variance of the aggregated feature vectors across all contributing views. To quantify the consistency of the observations contributing to this estimate, the weighted feature variance is computed as

$$\sigma_i^2 = \frac{1}{D} \sum_{d=1}^D (\mathbb{E}[F_d^2] - (\mathbb{E}[F_d])^2), \quad (17)$$

where D is the feature dimensionality and $\mathbb{E}[F_d]$ is the weighted mean of component d of the vector F_i defined in Eq. 15. High variance indicates disagreement between semantic observations from different viewpoints, suggesting that the Gaussian may lie near depth discontinuities, occlusions, or poorly reconstructed geometry. The uncertainty of Gaussian g_i is defined as $u_i = \sigma_i^2$.

Because different regions of the reef naturally exhibit varying levels of uncertainty, a local consistency measure was used rather than applying a global threshold. For each Gaussian g_i , a k -nearest neighbour search was performed in 3D space using euclidean distance, d , to obtain a local neighbourhood $\mathcal{N}_i$, where

$$d(P_i, P_{neighbor}) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2 + (z_j - z_i)^2}. \quad (18)$$

The uncertainty of the Gaussian was then compared to the neighbourhood statistics using a local z-score

$$z_i = \frac{u_i - \mu_{\mathcal{N}_i}}{\sigma_{\mathcal{N}_i} + \epsilon}, \quad (19)$$

where u_i is the uncertainty of the Gaussian, and $\mu_{\mathcal{N}_i}$ and $\sigma_{\mathcal{N}_i}$ are the mean and standard deviation of uncertainty values within the neighbourhood.

Only positive deviations are penalised, ensuring that points with lower uncertainty than their neighbours are not artificially boosted,

$$z_i = \max(0, z_i). \quad (20)$$

The resulting value is converted into a soft reliability weight using an exponential decay function

$$r_i = \exp(-\alpha z_i), \quad (21)$$

where α controls the strength of suppression. This formulation smoothly down-weights semantically inconsistent points while preserving the overall geometry of the reconstruction. To prevent excessive attenuation of sparse regions, the reliability weights are bounded such that $0.3 \leq r_i \leq 1$.

Finally, the aggregated semantic feature for each Gaussian is scaled by its reliability weight

$$F_i^{\text{final}} = r_i \cdot F_i. \quad (22)$$

This uncertainty-aware weighting suppresses noisy semantic projections without removing valid geometry, improving the stability of the resulting semantic field while preserving fine structural detail within the coral reef.

4.3.7 Dual-Mode Output: Decoupling Fidelity from Invariance

A fundamental challenge in creating a queryable ‘digital twin’ of a coral reef is balancing visual fidelity with semantic consistency. 3DGS achieves photorealism by representing view-dependent colour caused by lighting variations, specular reflections etc. using Spherical Harmonics (SH). The view dependent emitted colour C for a given Gaussian, observed from a viewing direction $\mathbf{v}$, is defined by the spherical harmonic series,

$$C(\mathbf{v}) = \sum_{l=0}^{l_{max}} \sum_{m=-l}^l c_l^m Y_l^m(\mathbf{v}). \quad (23)$$

Where l and m are the degree and order of the spherical harmonics, c_l^m are the learned harmonic coefficients, Y_l^m are the spherical harmonic basis functions and l_{max} is the maximum degree of the harmonics. This can be decomposed to isolate the base colour from the view-dependent reflections such that,

$$C(\mathbf{v}) = \underbrace{c_0^0 Y_0^0}_{\text{Base (DC)}} + \underbrace{\sum_{l=1}^{l_{max}} \sum_{m=-l}^l c_l^m Y_l^m(\mathbf{v})}_{\text{View-Dependent (Rest)}}. \quad (24)$$

In contrast to this, semantic identity is inherently view-invariant meaning a coral specimen remains the same species regardless of the observation angle. To resolve this conflict, the 3DGS pipeline was bifurcated into a visualisation mode and a semantic query mode.

For human-in-the-loop inspection and structural analysis, the renderer evaluates the entire summation of equation 24. The up to 45 higher-order terms modulate the base colour based on the camera’s viewing direction $\mathbf{v}$. These higher-order terms encode the complex, view-dependent lighting variability and water column effects characteristic of the underwater setting. Preserving these harmonics maximises the visual detail and photorealism of the reef, providing marine ecologists with an intuitive, highly detailed reconstruction of the habitat. Figure 7c shows a render of the 3D model with preserved higher-order harmonics, where a greater level of detail is observed, creating a more realistic digital twin when compared to Figure 7b, where the higher-order harmonics are zeroed.

Semantic identity, like our DINOv3 embeddings representing specific coral morphologies must be isotropic (view-invariant). If the feature vector $F(\mathbf{v})$ is extracted using the full SH series, the extracted embedding will artificially shift as $\mathbf{v}$ changes, degrading the accuracy of downstream tasks.

To enforce isotropy during semantic queries, the equation is collapsed by forcing all higher-order coefficients to zero such that

$$\forall l > 0, \quad c_l^m = 0. \quad (25)$$

Because the zeroth order basis function is a constant ($Y_0^0 = C_0 \approx 0.2821$), the view-dependent summation is entirely dropped. The extracted semantic feature F_i becomes strictly a function of the Direct Current (DC) coefficient f_{dc} where,

$$F_i = c_0^0 C_0 = f_{dc} \cdot C_0. \quad (26)$$

This mathematical decoupling ensures that the high-dimensional DINOv3 features projected into the 3D space remain stable and queryable from any arbitrary angle, without sacrificing the visual fidelity of the underlying model.

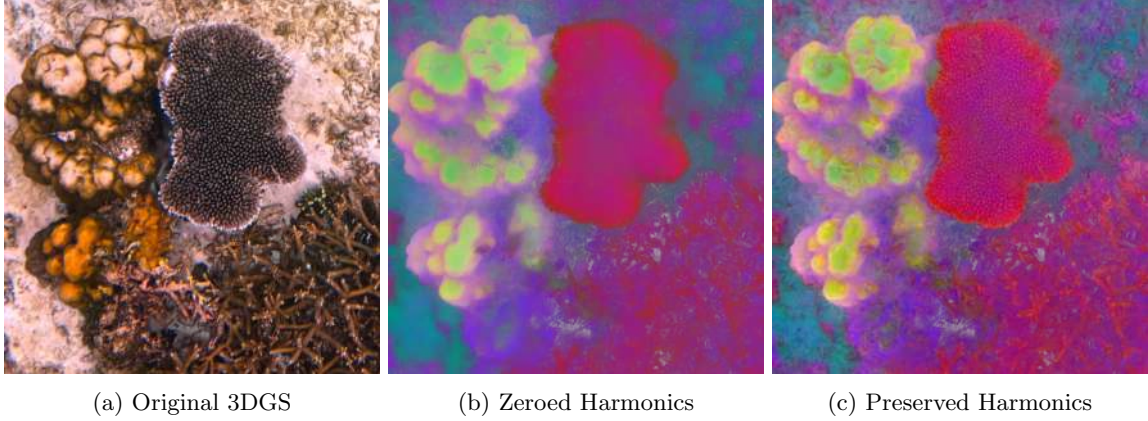


Figure 7: Demonstration of the improvement in clarity of intricate coral structures from preserving higher order spherical harmonics.

To visualise the high-dimensional semantic field, PCA is applied to the aggregated feature bank. The 64-dimensional feature vector F_i for each Gaussian is projected onto its three primary principal components, which are then normalised and mapped to the RGB colour channels. This transformation allows for a direct visual interpretation of the semantic clusters within the 3D reconstruction, where colour proximity indicates semantic similarity as seen in Figures 7b and 7c.

4.4 Interactive Semantic Querying

While PCA projection provides a global qualitative overview of the semantic field, it inherently compresses 64 dimensions of data into three, potentially masking subtle taxonomic or health-related distinctions. To leverage the full depth of the DINOv3 embeddings, an interactive semantic search tool was developed.

This tool bridges the 2D and 3D domains by allowing a user to select a specific feature of interest within a source RGB image. Given a pixel coordinate (u, v) in image j , the corresponding high-dimensional feature vector F_{ij} is sampled. This vector serves as a semantic query vector, Q_{ij} that is then compared against the entire 3D feature bank F using cosine similarity (as in Eq. 4).

The resulting similarity scores $S_i \in [-1, 1]$, which represent the cosine similarity between the centred 2D query and the centred 3D feature bank, are normalized and mapped to a divergent colour-map. These colour values are directly injected into the spherical harmonic DC coefficients (f_{dc}) of the model. This enables the 3D visualisation of all semantically similar structures across the reef, significantly reducing the manual effort required for traditional 3D segmentation or dense point-cloud labelling. The 3D model is effectively recoloured to act as a semantic heatmap, highlighting regions that share the latent feature signature of the selected query point.

Beyond qualitative visualisation, this approach facilitates automated, zero-shot 3D segmentation. By applying a binary threshold τ to the similarity scores S_i , the system can systematically filter the 3D scene to isolate specific semantic classes. Formally, a Gaussian point P_i is retained in the filtered subset

$\mathcal{P}_{seg}$ if

$$\mathcal{P}_{seg} = \{P_i \mid S_i \geq \tau\}. \quad (27)$$

This thresholding mechanism allows researchers to mask out background elements such as sand, rubble, or water column noise leaving only the target coral morphologies. As implemented in the data pipeline, this filtering operation effectively prunes the geometry to generate a new, semantically pure sub-model. Depending on the feature quality, this allows for the isolation of the specific target biomass, streamlining downstream tasks such as volumetric estimation and structural complexity analysis without requiring exhaustive manual annotation.

4.5 Evaluation Methodology

4.5.1 Split-Half Consistency

To quantify the reliability of the multi-view feature aggregation, a split-half consistency test was conducted. The set of source images contributing to the 3D projection was split randomly into two disjoint subsets, yielding two halves of approximately equal size. (276 and 279 matched images respectively). The full projection pipeline (Section 4.3) was executed independently for each half, producing two separate 64-dimensional feature vectors F_i^A and F_i^B for every Gaussian g_i . To reduce sampling bias, this procedure was repeated across multiple random splits with different seeds, and results were aggregated across runs.

The agreement between the two independent estimates was measured using per-Gaussian cosine similarity,

$$S_i = \frac{F_i^A \cdot F_i^B}{\|F_i^A\| \|F_i^B\|}. \quad (28)$$

Only Gaussians observed by a minimum of three views in both halves were included in the evaluation (100% of Gaussians for this dataset). This test is independent of ground-truth labels and it instead directly measures the internal consistency of the pipeline. Therefore, a high mean similarity indicates that the resulting semantic representation is robust to the specific subset of source views, so would generalise to repeated surveys of the same site.

4.5.2 3D Cluster Analysis

For the 3D semantic models, k-means clustering was applied to the full projected feature banks to evaluate whether the multi-view aggregated embeddings retain meaningful semantic structure. The same centring and L2-normalisation preprocessing was applied to ensure consistency with the 2D analysis.

Prior to clustering, a geometry-based quality filter was applied to exclude Gaussians unlikely to represent valid reef surface. Gaussians with post-sigmoid opacity below 0.1 were removed, as these semi-transparent primitives typically model volumetric effects rather than solid geometry and carry unreliable semantic features. Additionally, spatial outliers were identified using the Median Absolute Deviation (MAD); points deviating by more than $2.5\hat{\sigma}$ from the median position on any axis were excluded as reconstruction floaters.

Due to the scale of the filtered feature bank ($N > 2.5 \times 10^6$), MiniBatchKMeans was used for computational tractability, fitted on a stratified subsample of 500,000 Gaussians with full-dataset label assignment via prediction. Clustering was performed for $k \in \{2, 3, 4, 5, 6, 7, 8\}$ to characterise how cluster quality varies with semantic granularity. Cluster quality was evaluated using the silhouette coefficient [40], computed on a random subsample of 100,000 Gaussians using cosine distance to maintain consistency with the centring and normalisation preprocessing. To enable visual inspection, cluster labels were mapped to distinct colours and injected into the spherical harmonic DC coefficients of the model.

4.5.3 3D Validation Labelling Protocol

To provide direct quantitative evaluation of the 3D semantic field, a sparse point-labelling regime was conducted on the Tabuhan P1 source imagery. Images were filtered to those with significant coverage of the $5 \times 5\text{m}$ sub-block using COLMAP point visibility counts. Random points were generated within the central 50% of each image to avoid edge distortion artifacts leading to labelling error, and shuffled across images to reduce labelling drift.

Each point was classified into one of seven classes based on the substrate visible directly under the randomly selected crosshair as demonstrated in Figure 8.

The ‘unknown’ class was used liberally for any point where confident identification was not possible, including ambiguous boundaries and occluded points. Certain class boundaries, notably rock versus other life, remained challenging even with this protocol. Labelling was performed by the author and supervisors, none of whom are trained taxonomists, a limitation that is acknowledged and discussed in Section 6. This motivates the deliberately coarse class taxonomy, which avoids species-level distinctions that would require expert knowledge to apply consistently.

Two separate sets of 300 point labels were annotated independently by three annotators following the identical protocol, with no access to each other’s labels. Because ‘unknown’ does not represent a true substrate class, all agreement metrics were computed after excluding these labels. This ensures that the resulting metrics specifically reflect the taxonomic reliability of the identified substrates rather than agreement on image ambiguity. Any point where any annotator assigned ‘unknown’ were therefore excluded.

Overall multi-annotator agreement was calculated using Fleiss’ Kappa (κ) [41],

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \quad (29)$$

where $\bar{P}$ is the mean observed proportion of agreeing annotator pairs across all items and $\bar{P}_e$ is the expected agreement by chance. As a result, Fleiss’ Kappa corrects for chance agreement between all three annotators.

To assess agreement of individual classes, Fleiss’ Kappa was computed for each class independently. For a given class c , each annotator’s labels were treated as either c or not c , and Fleiss’ kappa was then applied to the binary labels across all annotators.

The resulting agreement metrics validate the consistency of the labelling protocol, and provide a human performance baseline against which segmentation performance can be contextualised, since if the reference labels are inaccurate the model cannot correctly identify them.

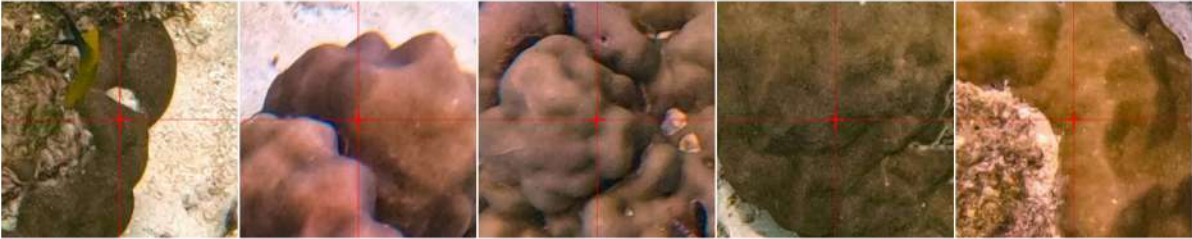
4.5.4 3D Segmentation Validation

The labels from Section 4.5.3 were used to evaluate the models segmentation performance at three levels of class granularity. A binary evaluation collapses all coral classes into a single coral class against all non-coral substrates, testing the model’s ability to differentiate living morphology from background. An intermediate substrate test collapses sand and rock into a single ‘substrate’ class to reflect the boundary ambiguity between these two categories explained in Section 5.2.6, this is compared against all other classes uniquely. A complete evaluation retains all classes individually, assessing whether the feature space captures morphological distinctions within the coral category.

Classes that either occurred too rarely to support evaluation (equipment) or resulted in per-class Fleiss’ κ below the substantial agreement threshold of 0.6 (other living, $\kappa = 0.19$) were excluded from all granularities. This is because points that annotators could not reliably agree on do not provide a meaningful test result.

Because the same points were labelled by multiple annotators, a single ground-truth label was decided per point using majority voting across annotators. Points where no majority existed were excluded. The evaluation was also repeated using a strict consensus subset, where only labels with complete agreement

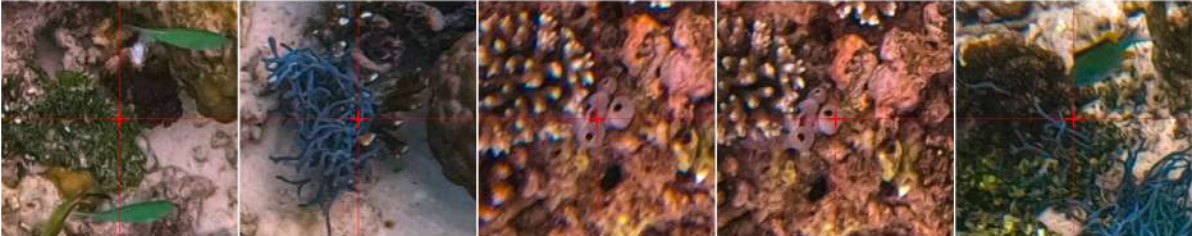
Coral (Smooth)



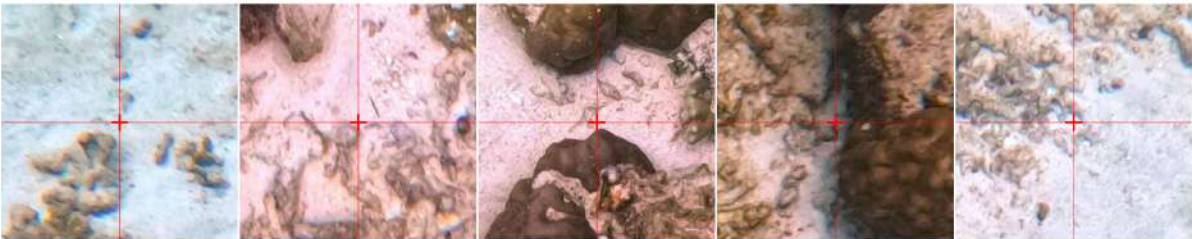
Coral (Branching)



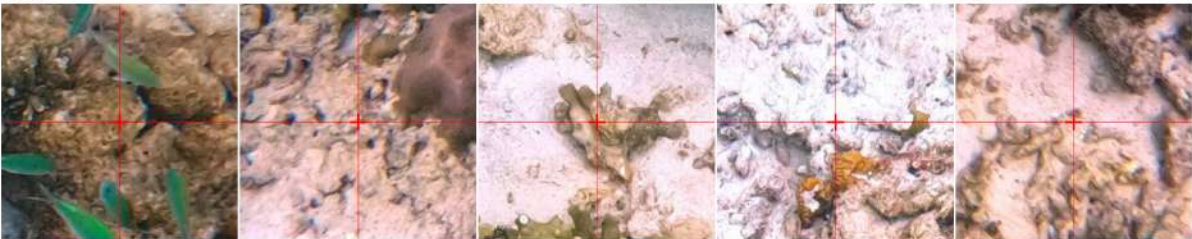
Other Living



Sand



rock



unknown

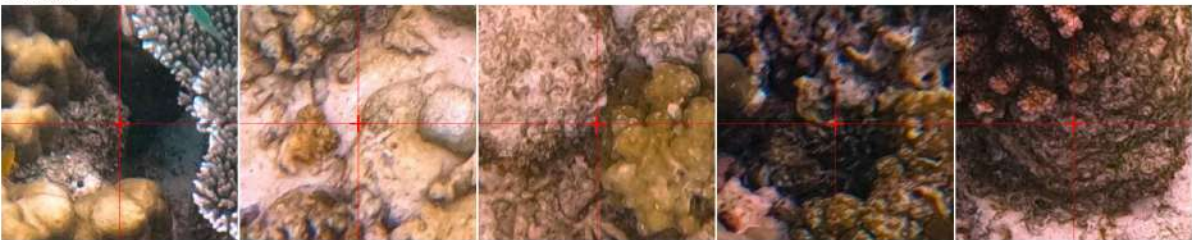


Figure 8: Examples of classes of manually labelled points from the Tabuhan P1 source imagery. Labels are assigned based on the surface beneath the red crosshair.

were included to provide an upper-bound estimate of performance on unambiguous points. For robustness, per-annotator results are also evaluated to ensure results aren't corrupted by a single annotator being a distinct outlier.

In all cases, each labelled 2D point is projected into the 3D model using the camera geometry described in Section 4.3.1 to identify the nearest Gaussian primitive, and the associated 64-dimensional feature vector was extracted from the feature bank.

Few-shot classification using these features was then performed following the prototype-based cosine similarity protocol established for the 2D evaluation (Section 4.2.2). Randomly labelled 2D points were selected as references, and the mean of the corresponding 3D features is taken to form a class prototype. All remaining points are classified by cosine similarity of their corresponding 3D feature to the class prototypes. To account for sensitivity to reference selection, the evaluation was repeated across 100 random seeds and performance is reported as the mean and standard deviation of the weighted F1-score. This was conducted for $k \in \{1, 3, 5, 10\}$ to demonstrate how performance scales with label count.

To contextualise the model's performance against the reliability of the labels themselves, a human performance ceiling is computed for each granularity. This represents the weighted F1 an average annotator would achieve if scored against another annotator's label, and therefore bounds the accuracy any classifier can reach with the labels provided.

The ceiling is computed on the subset of points labelled by all three annotators P_{total} (not labelled unknown), ensuring consistent comparisons across pairs. For a class mapping. For each annotator pair (a, b) , their agreement is

$$A_{ab} = \frac{1}{P_{total}} \sum_{i \in P_{total}} 1[l_i^a = l_i^b], \quad (30)$$

where l_i^a is the assigned label for point i by annotator a . The ceiling is the mean across the three unordered annotator pairs (j, k) such that

$$C = \frac{1}{3} \sum_{(j, k)} A_{jk}. \quad (31)$$

This is computed once per granularity. The ceiling is trivially 1.0 under consensus labels, where only unanimous points enter the evaluation.

4.5.5 Pipeline Ablation

To quantify the cumulative contribution of the projection pipeline's engineering components, a two-condition ablation was conducted. The naive baseline performs bilinear-only upsampling, no occlusion culling, uniform averaging across views and no reliability filtering. The full pipeline enables all five components as described in Sections 4.3.3 - 4.3.6. Both conditions project onto identical geometry using the same source features and are evaluated using the same quality-filtered Gaussian subset.

Three tests are ran using both the naive baseline and full pipeline for comparison. A split-half consistency test as described in Section 4.5.1, where source images are randomly partitioned into two disjoint halves five times and aggregated across all five splits to ensure reliability of results. Silhouette scores at $k \in \{2, 4, 8\}$ are calculated for both projection methods to assess cluster separability in the projected feature space. Finally, few-shot classification as described in Section 4.5.4 is performed using both the naive projection features and full pipeline features, yielding weighted F1 at $k \in \{1, 3, 5, 10\}$ under majority-vote label resolution as this is the most representative for realistic usage.

5 Results

The results are presented in two stages, mirroring the structure of the methodology. First, the 2D validation results assess whether DINOv3 embeddings capture semantically meaningful structure in underwater coral imagery, characterising the qualitative feature quality, computational trade-offs, and quantitative clustering and few-shot classification performance of both architectures. Second, the 3D results evaluate whether these semantics are faithfully preserved when projected into the Gaussian Splatting representation, assessed through qualitative inspection of the semantic field, split-half consistency analysis, unsupervised cluster separability, few-shot classification, and a pipeline ablation study. Together these two stages address the core hypothesis that high-dimensional DINOv3 embeddings can be lifted into an explicit 3DGS representation to support discriminative, label-efficient 3D semantic analysis of coral reefs.

5.1 2D Results: Validating Domain Efficacy of DINOv3

The 2D validation stage aims to confirm that DINOv3 embeddings capture semantically meaningful structure in underwater coral imagery without domain-specific fine-tuning. First, qualitative analysis of the PCA feature maps assesses the spatial coherence and semantic separability of both architectures. Second, clustering experiments on the Moorea Labelled Corals dataset quantify the discriminative quality of the embeddings under supervised and unsupervised conditions. Third, few-shot classification results characterise performance under realistic sparse-label constraints. Together establishing the semantic quality of the 2D feature space that underpins the subsequent 3D projection.

5.1.1 Qualitative Analysis of Latent Features

As illustrated in Figure 9 both architectures demonstrate an intrinsic ability to identify semantically similar reef elements without domain-specific fine-tuning. The first three principal components of the high-dimensional feature space are mapped to the RGB channels resulting in qualitatively distinct, colour-coded clusters. Variance of these qualitative clusters match with biological and structural categories. For example, in Figure 9a, the green colour maps to all examples of a particular distinct coral type.

A significant performance gap is visible in the spatial fidelity of the two models. The ViT-7b output (Figure 9a) is notably sharper and more distinct than the ConvNeXT-L output.

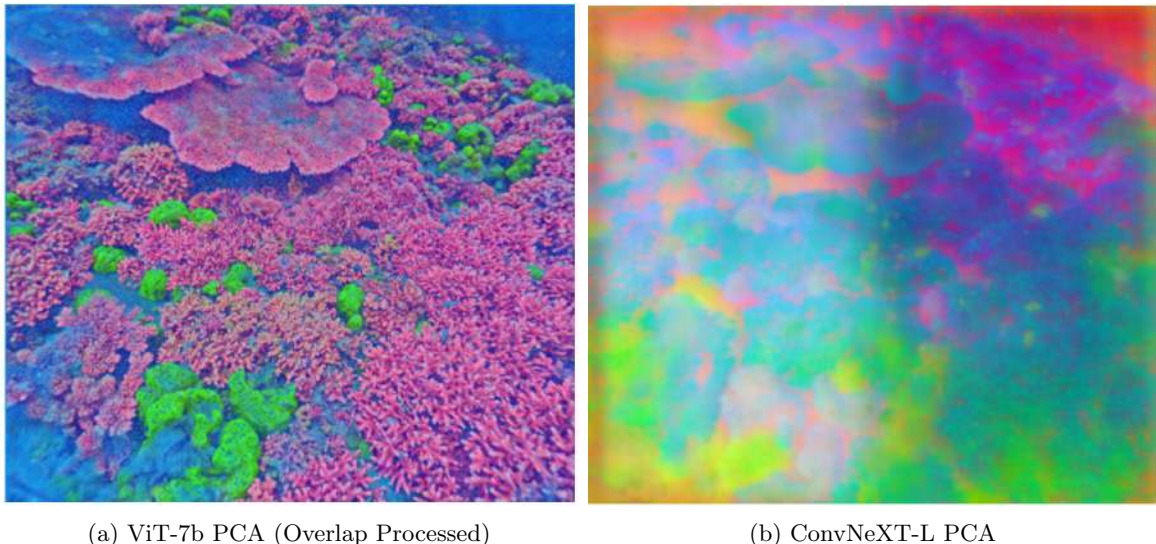


Figure 9: Comparison of feature quality of ViT-7b (a) & ConvNeXT-L (b) PCA outputs from source image (Figure 5a). The finer detail in (a) is due to its higher feature resolution 335×293 , compared to (b) with 167×146 . Both are upsampled to source image resolution for visualisation.

The overlapping strategy detailed in Section 4.2.1 successfully eliminated the seam artifacts seen in the

raw ViT processing and the architecture’s inherent advantages are clear:

- **Spatial Resolution:** The ViT utilizes a patch size (stride) of 16, resulting in a feature resolution of 335×293 . In contrast, the ConvNeXT’s stride of 32 yields a 167×146 map, leading to the interpolation-induced blurring seen in Figure 9b.
- **Embedding Depth:** The ViT’s embedding dimension is approximately 2.6 times larger than the ConvNeXT-L ($D = 4096$ vs 1536). This higher dimensionality allows for the capture of more nuanced visual variance, which is critical for distinguishing between similar coral species.

Despite the superior fidelity of the ViT-7b, the computational cost is substantial. Using a 16GB VRAM GPU, the sliding-window ViT pipeline incurred approximately $60\times$ longer processing times per image than the ConvNeXT-L.

5.1.2 Clustering & Few Shot Results

The results of both methods of clustering (Figure 10), show that the models are qualitatively able to cluster the image correctly, further demonstrating their validity for the underwater domain. In the supervised kNN results (Figure 10b/d), the filtering methodology is largely successful, prior results classified the entire tape measure as sand for example. Whereas, large portions of the monitoring equipment are now correctly classified as ‘unknown’, as intended, due to the lack of consistent labels for these ‘off’ items. A key performance difference is also seen between the ConvNeXT model and the ViT. The ConvNeXT model struggles to resolve the thin (X-shaped) string across the image due to its lower feature resolution, this contrasts with the ViT’s ability to maintain high-frequency boundaries. This suggests that while both models are effective for broad substrate classification, the ViT is uniquely suited for monitoring structurally complex organisms where fine-scale geometric precision is a prerequisite for accurate biomass estimation.

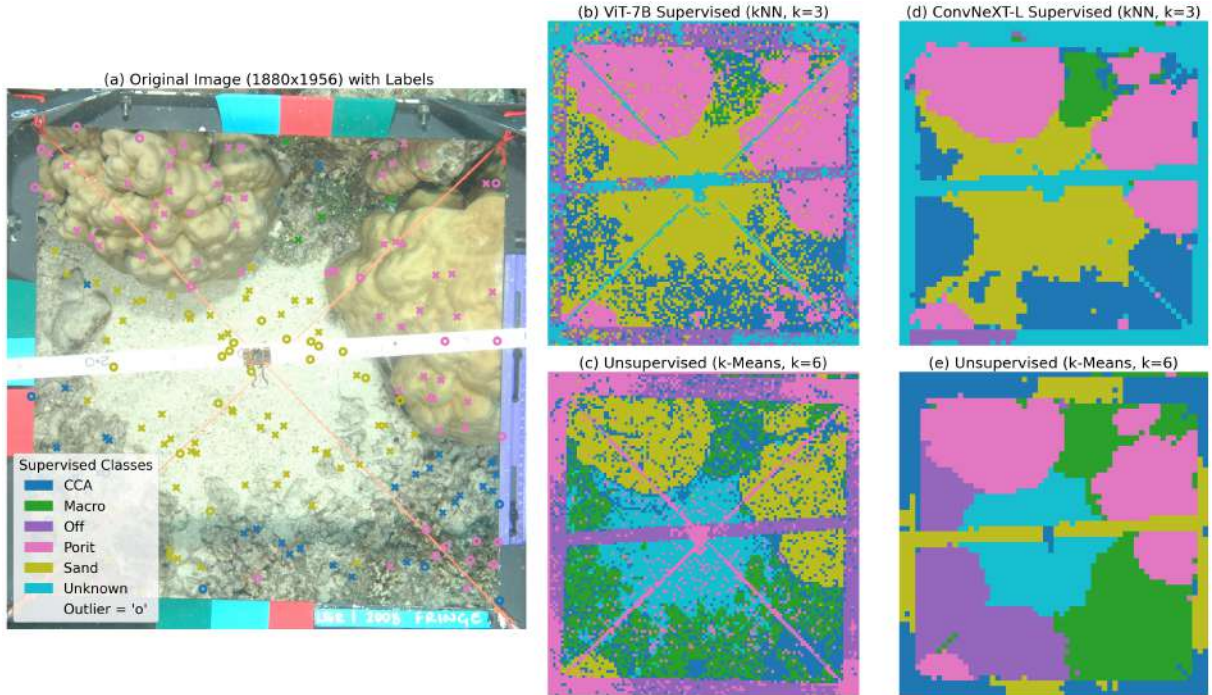


Figure 10: KNN & k-means Clustering Comparison for ViT & ConvNeXT for labelled image (a) (where x’s are included labelled points and o’s excluded). Showing segmentation masks for the ViT (b & c) and the ConvNeXT (d & e). The top row shows supervised results with mask colours corresponding to the legend. The bottom row shows unsupervised results with random class colours.

The unsupervised results (Figure 10c/e) are also strong, indicating that the embeddings are able to accurately capture key differences in coral species, sand etc. with no additional guidance. The choice

of $k = 6$ was based on the number of classes in the labelled dataset for this particular image. Testing varying k and using silhouette scores revealed a strong bias for $k = 2$. This bias for $k = 2$ underscores the models’ ability to autonomously differentiate between natural reef substrate and anthropogenic materials (e.g., monitoring equipment and tape measures). While this limited the direct evaluation of coral sub-species segmentation in an unsupervised setting, it validates the robustness of the DINOv3 embeddings for automated ‘clean-up’ of reef datasets prior to ecological analysis. While this demonstrates that the model effectively identifies key semantic differences, it was not useful for testing of the coral segmentation efficacy.

Finally to test the overall aim of using these foundational models in a sparsely labelled setting, their few-shot classification capability was tested. Using 500 images from the Moorea Labelled Corals dataset, but only providing k labelled reference points for each class per image. KNN clustering using $k = 5$ reference points was performed and the predictions were tested against remaining labelled pixels. This resulted in a weighted average F1-score of 0.58 and 0.56 for the ViT and ConvNeXT respectively. The F1 scores across $k \in \{5, 10, 25\}$ for ViT-7b are presented in Table 1. This performance demonstrates that in a highly data-sparse scenario, the DINOv3 features are clearly separable.

To contextualise DINOv3 performance against the closest comparable baseline, Table 1 presents mIoU alongside the sparse-label results of Raine et al. [11]. Direct numerical comparison is limited by several factors, including differing class counts which is discussed further in Section 6.2. As such, these results should not be interpreted as confirmation of superior performance. However, they do demonstrate that DINOv3 segmentation performance in a sparse-label setting is comparable to established baselines.

Table 1: 2D few-shot classification performance on the Moorea Labelled Corals dataset (DINOv3 ViT-7b) alongside the sparse-label DINOv2 results of Raine et al. [11] (who do not report F1 scores) on the UCSD Mosaics dataset. Raine et al. use denoised DINOv2 features across 33 classes, our results use raw DINOv3 features across 13 classes. Both use k reference points.

Method	mIoU $\uparrow$			Weighted F1 $\uparrow$		
	$k=5$	$k=10$	$k=25$	$k=5$	$k=10$	$k=25$
Denoised DINOv2 [11]	0.32	0.43	0.58	–	–	–
DINOv3 ViT-7b (ours)	0.41	0.50	0.74	0.58	0.66	0.83

5.2 3D Results

The 3D results stage addresses whether the semantic quality established in the previous section is preserved through the projection pipeline and whether the resulting 3D feature field supports meaningful ecological analysis. First, Qualitative inspection and split-half consistency analysis assess the spatial coherence and reliability of the multi-view aggregation. Subsequently, unsupervised cluster separability and few-shot classification evaluate whether the discriminative capability of the projected features is preserved in 3D. A pipeline ablation study then characterises the impact of the projection pipeline’s aggregation strategy on feature quality and consistency.

5.2.1 Qualitative Assessment of the Semantic Field

Injecting the three principal components of the aggregated 64-dimensional feature bank as RGB colour channels into the original model produced semantically coherent 3D reconstructions for both architectures, as displayed in Figure 11.

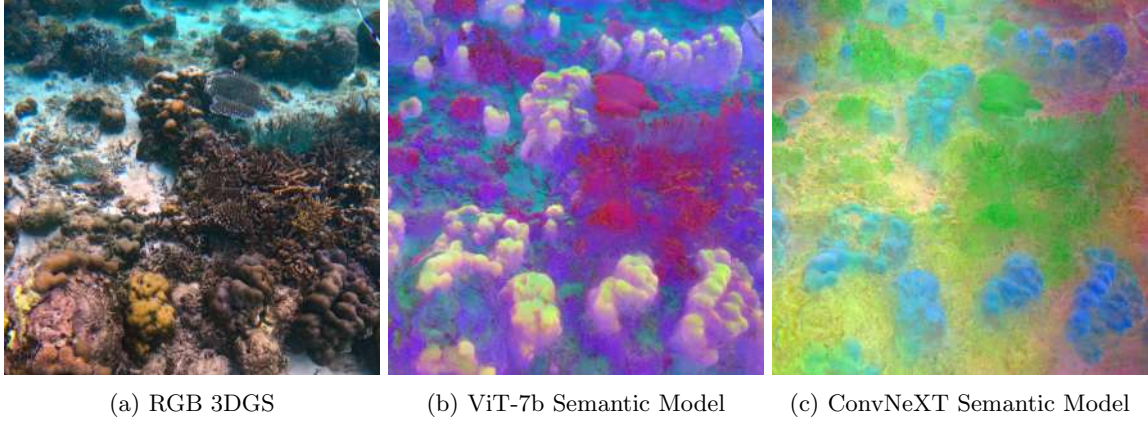


Figure 11: Example view of standard RGB 3DGS model (a) alongside semantic splatting models using ViT (b) and ConvNeXT (c).

A primary advantage of the multi-view aggregation strategy is the emergence of spatial coherence across complex geometries. Because each Gaussian’s feature vector F_i is computed as a confidence-weighted average across all valid camera views (Eq. 15), transient noise and per-frame lighting artifacts are effectively regularised. As seen in Figure 11, the semantic colouring of coral colonies remains consistent across occlusions and high-curvature surfaces, providing empirical evidence that DINOv3 features are viewpoint-invariant within the reef domain. This regularisation effect is a direct consequence of the 3D integration and would not be achievable through independent 2D frame analysis alone.

5.2.2 Architecture Comparison in 3D

A notable finding of the 3D projection is the convergence in boundary preservation between the two architectures. In the 2D analysis (Section 5.1), the ViT-7b demonstrated markedly sharper boundary preservation than the ConvNeXT-L, attributable to its finer 16×16 patch stride. However, in the 3D semantic models (Figure 12b vs. Figure 12c), the edge preservation at the interface between coral morphology and sand substrate appears highly similar, with comparable levels of semantic bleeding observed for both models.

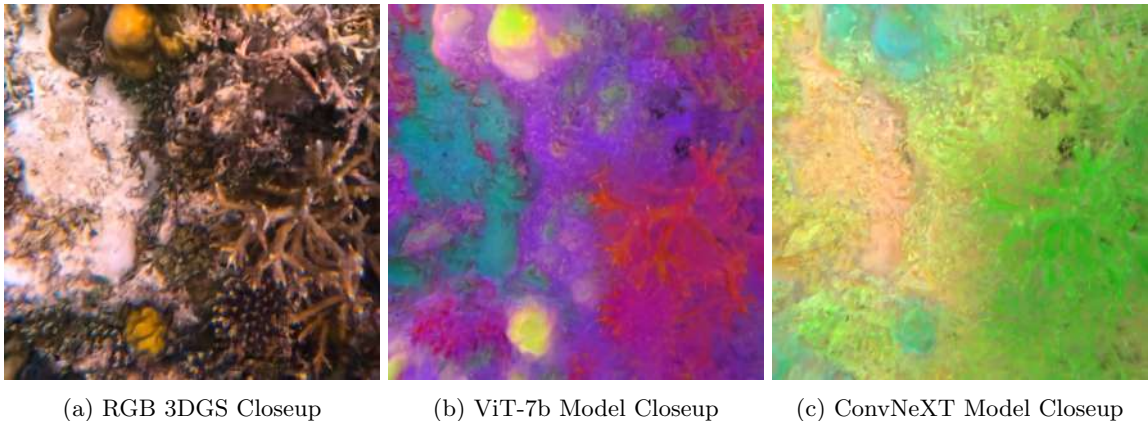


Figure 12: Close-up of a coral-sand boundary comparing the RGB 3DGS model (a) with semantic models from the ViT-7b (b) and ConvNeXT-L (c). Despite the ViT’s superior 2D resolution, both architectures exhibit comparable edge preservation at substrate interfaces following multi-view aggregation.

This convergence can be attributed to two combined factors. The guided filter upsampling applied to both feature maps prior to projection uses the high-resolution RGB image as an edge-preserving guide (Section 4.3.3), directly compensating for the ConvNeXT’s coarser native resolution by re-aligning feature boundaries with intensity gradients in the source imagery. Additionally, the multi-view weighted averaging provides additional sharpening. Gaussians at semantic boundaries receive conflicting obser-

variations across viewpoints, increasing their semantic variance and thus reducing their reliability weight (Section 4.3.6). This soft suppression of uncertain boundary Gaussians produces cleaner transitions in both models, reducing the architecture-dependent gap observed in 2D.

However, two further qualitative differences emerge that favour the ViT architecture for semantic fidelity. First, the ConvNeXT model exhibits reduced inter-class feature separation. As seen in Figure 11c, the PCA colouring of sand substrate appears notably closer in hue to adjacent coral colonies than in the ViT output (Figure 11b), where these classes are clearly distinguished. This compression of the semantic distance between classes reduces the discriminative resolution available for downstream tasks such as thresholded segmentation or few-shot classification.

Second, the ConvNeXT output displays visible banding artifacts (Figure 13b), where the PCA colouring is locally coherent within spatial bands but varies across the model in a pattern inconsistent with the underlying reef structure. In contrast, the ViT output (Figure 13a) displays globally consistent, view invariant features.

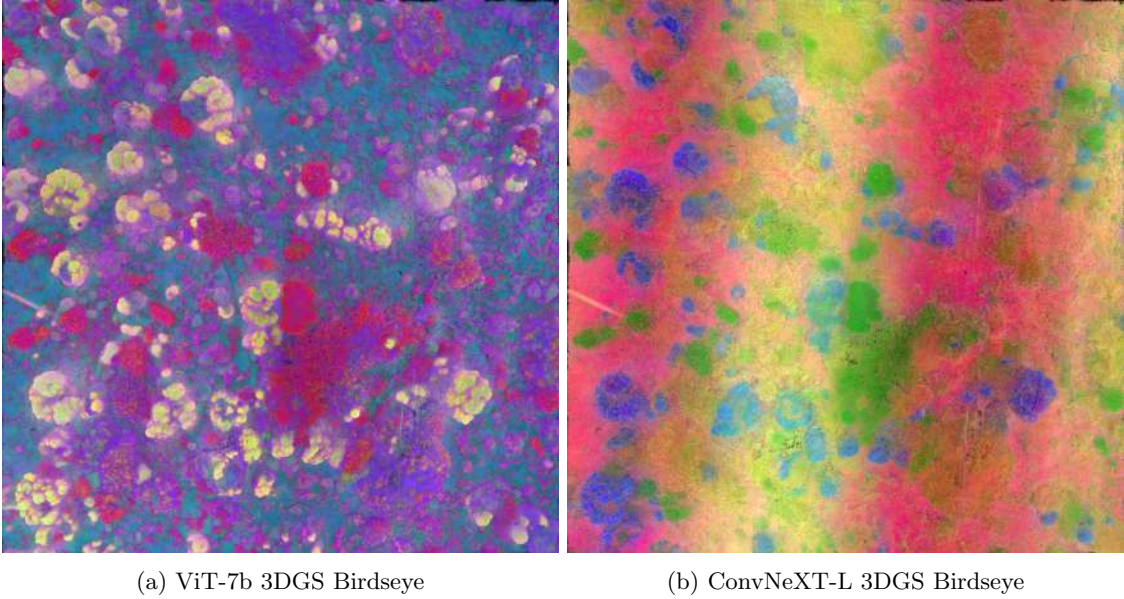


Figure 13: Bird’s-eye comparison of 3D semantic models. The ViT-7b model (a) exhibits globally consistent PCA colouring, whereas the ConvNeXT-L model (b) displays spatial banding where feature hue shifts systematically across the scene, inconsistent with the underlying reef morphology.

The observed banding in Figure 13b is attributed to the architectural sensitivity of convolutional networks to local illumination gradients. In the underwater setting, systematic lighting variation arises from camera altitude, survey trajectory, and water column attenuation. The hypothesis is that ConvNeXT’s locally-bounded receptive field encodes these intensity gradients directly into the feature representation, whereas the ViT’s global self-attention mechanism provides implicit normalisation by conditioning each patch token on the full image context. When features from multiple views are aggregated, these illumination-correlated components do not fully cancel because cameras along a survey transect share similar lighting geometry, producing the observed coherent bands.

These two effects are likely related, since if a portion of the ConvNeXT’s feature variance is consumed by encoding illumination rather than semantic identity, PCA will allocate principal components partially to this nuisance variation, compressing the apparent distance between genuine taxonomic classes. This interpretation is consistent with the lower explained variance at the PCA elbow for the ConvNeXT (Figure 6b), which suggests a more diffuse variance structure where semantic and photometric signals are less cleanly separated.

5.2.3 Split-Half Consistency

Following the protocol described in Section 4.5.1, the split-half test was performed for both the ViT-7b and ConvNeXT-L architectures to assess whether the 3D feature field converges to a stable, view-invariant representation. All 2,803,311 Gaussians in the model satisfied the minimum view threshold in both halves, confirming high observational redundancy of the survey coverage. A random split was used specifically to avoid conflating view-invariance with illumination consistency, as sequential COLMAP ordering correlates with camera trajectory and would preserve similar lighting geometry across splits. This avoids artificially preserving the observed banding effects seen in Figure 13b. The experiment was repeated across five independent random splits to quantify sensitivity to the specific split selected, with the results summarised in Table 2.

Table 2: Split-half consistency results under random image partitioning, aggregated across five independent runs. Per-Gaussian cosine similarity is computed between feature banks constructed from two disjoint halves of the source image set.

Metric	ViT-7b	ConvNeXT-L
Mean similarity $\uparrow$ (5 runs)	0.899 ± 0.006	0.874 ± 0.010
Median similarity $\uparrow$	0.926	0.900
Standard deviation $\downarrow$	0.091	0.096
% above 0.9 $\uparrow$	63.8	49.8
5th percentile $\uparrow$	0.721	0.684

Both architectures achieve strong consistency metrics across all five splits, with mean cosine similarities of 0.899 ± 0.006 and 0.874 ± 0.010 for ViT-7b and ConvNeXT-L respectively. This indicates that the projected semantic features are highly stable under halved input data. The 5th percentile remaining above 0.68 in both cases demonstrates that even the least consistent Gaussians maintain substantial agreement across independent reconstructions. The low variance across runs for both models, particularly ViT-7b, indicates that these results are not sensitive to the specific split used.

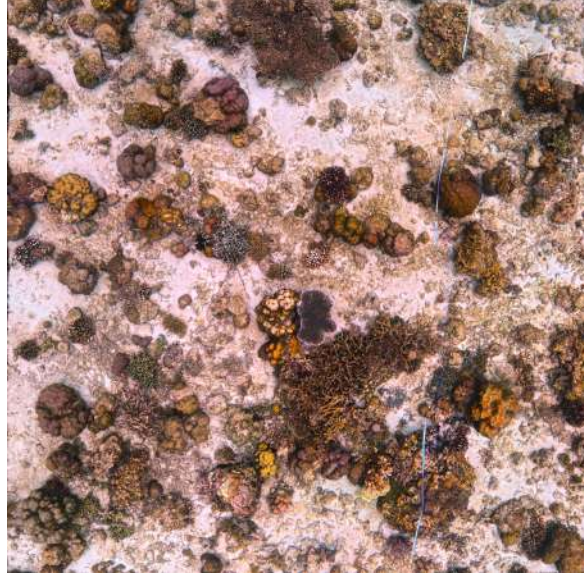
ViT-7b outperforms ConvNeXT-L across every metric, consistent with its higher feature resolution and embedding depth. ConvNeXT-L additionally exhibits greater run-to-run variance (0.010 vs 0.006), suggesting its representations are more sensitive to the lighting distribution of each partition. This is consistent with the banding identified in Section 5.2.2, where ConvNeXT features encode illumination-correlated structure that is not view-invariant. Together, these results confirm that the aggregation pipeline reliably produces stable 3D representations from independent image subsets, with ViT-7b providing the more consistent semantic field.

The split-half test establishes pipeline reliability but does not assess whether the stable representations encode semantically meaningful structure. This distinction motivates the cluster separability analysis in Section 5.2.4, which evaluates discriminability directly. The two tests therefore characterise the distinct properties of reliability and discriminability respectively for the semantic field.

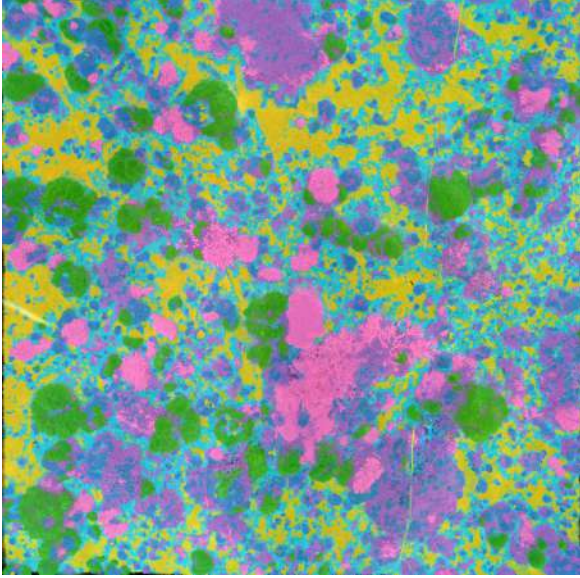
5.2.4 Cluster Separability

To assess whether the projected features encode semantically meaningful structure, unsupervised k-means clustering was applied to the 3D feature banks following the protocol described in Section 4.5.2. Prior to clustering, a geometry-based quality filter removed Gaussians with post-sigmoid opacity below 0.1 and spatial outliers beyond $2.5\hat{\sigma}$ from the median position, retaining 2,510,993 of 2,803,311 Gaussians (89.6%).

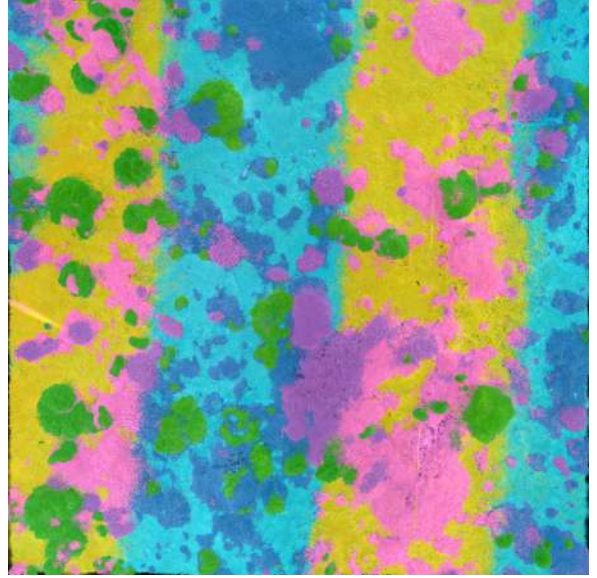
At $k = 6$, the cluster assignments were injected into the 3DGS model as false-colour labels for direct visual inspection. Figure 14 presents the resulting models for both architectures alongside the original RGB reconstruction.



(a) RGB 3DGS Reference



(b) ViT-7b ($k = 6$)



(c) ConvNeXT-L ($k = 6$)

Figure 14: Unsupervised k-means clustering ($k = 6$). The ViT-7b clusters (b) align with visible ecological boundaries in the RGB reference (a). In contrast, two of the six ConvNeXT-L clusters (c) correspond to systematic illumination bands rather than reef taxonomy.

The ViT-7b clusters (Figure 14b) align closely with ecologically meaningful boundaries visible in the RGB reference, with distinct clusters corresponding to identifiable substrate types and coral morphologies. As hypothesised in Section 5.2.2, the ConvNeXT-L output (Figure 14c) reveals that two of its six clusters correspond to spatial illumination bands rather than ecological categories, confirming that the lighting issues persist through to the clustering stage.

The quantitative cluster metrics, summarised in Table 3, support and extend this observation.

Table 3: Silhouette scores after geometry-based quality filtering ($N = 2,510,993$).

k	ViT-7b $\uparrow$	ConvNeXT-L $\uparrow$
2	0.292	0.251
4	0.302	0.269
6	0.306	0.239
8	0.288	0.233

The ViT-7b achieves higher silhouette scores across all values of k , with the gap widening from $\Delta = 0.041$ at $k = 2$ to $\Delta = 0.067$ at $k = 6$. The two architectures peak at different resolutions, the ViT-7b at $k = 6$ and the ConvNeXT-L at $k = 4$, suggesting that the ConvNeXT’s feature space supports coarse partitions but fragments under finer taxonomic demands. While the absolute silhouette scores are moderate (peak of 0.306), this is expected for completely unsupervised clustering of continuous ecological data, where semantic categories exhibit gradual spatial and spectral transitions rather than discrete boundaries.

The illumination-band clusters visible in Figure 14c reveal that semantic identity and illumination context are entangled in the ConvNeXT embedding space, representing a fundamental limitation of the architecture’s feature representation. Gaussians belonging to the same ecological class but observed under different lighting conditions (due to camera altitude or position along the survey transect) receive divergent feature vectors. This entanglement forces k-means to choose between grouping by taxonomy, which incurs high intra-cluster variance from the lighting gradient, and grouping by illumination, which fractures genuine semantic classes across multiple clusters. In practice both effects occur simultaneously, contributing to the ConvNeXT-L’s lower silhouette scores across all values of k . However, illumination entanglement is not the sole factor. The ConvNeXT’s coarser spatial resolution (stride 32 versus 16) and lower embedding dimensionality ($D = 1536$ versus 4096) also reduce boundary precision and taxonomic nuance respectively. At $k = 6$, the lighting bands in Figure 14c provide direct evidence of illumination contamination, but the broader performance gap across all values of k reflects the cumulative effect of these architectural differences. Even at $k = 2$ (Figure 15), the ViT-7b achieves a notably higher silhouette score than the ConvNeXT-L (0.292 vs 0.251), indicating that the feature degradation extends to the coarsest coral-versus-substrate partition.

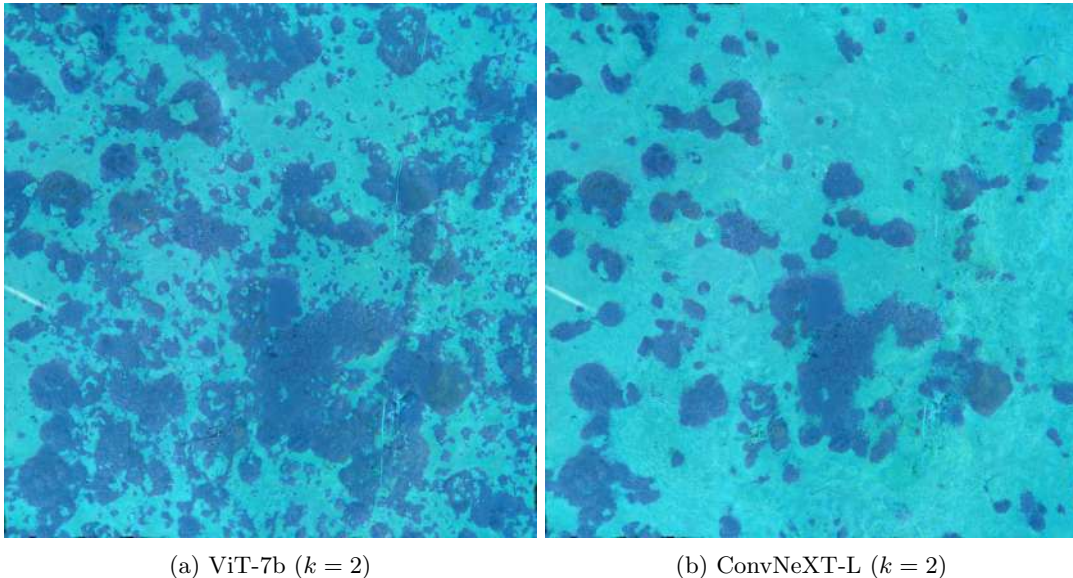


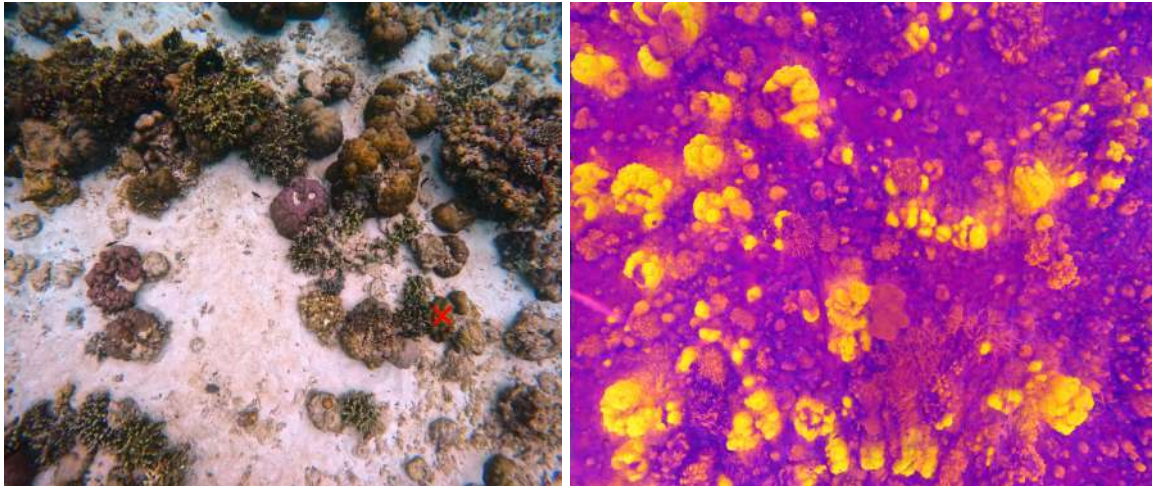
Figure 15: Binary clustering ($k = 2$). The ViT-7b (a) captures a comprehensive coral-versus-substrate partition. The ConvNeXT-L (b) systematically under-represents coral coverage, assigning substantial coral regions to the substrate class. Raw RGB reference 3DGS model and equivalent views of $k=6$ clustered models are available in Figure 14 for comparison

As seen in Figure 15, the ViT-7b produces a comprehensive binary partition where coral coverage is well-delineated from the surrounding substrate. The ConvNeXT-L, by contrast, systematically under-represents coral extent, assigning substantial regions of live coral to the substrate class.

Combined with the split-half consistency results (Section 5.2.3), a clear characterisation of the two architectures emerges. Both produce reliable semantic representations (mean cosine similarity > 0.87), but the ViT-7b is more reliable (0.899) and encodes substantially greater discriminative structure. The illumination entanglement in the ConvNeXT’s features degrades cluster quality across all tested values of k , including the coarsest binary partition, indicating that the limitation is fundamental to the representation rather than confined to fine-grained analysis. The ConvNeXT-L’s primary advantage is its $\sim 60\times$ faster processing time, which may justify its use in time-critical scenarios where rapid reconstruction is prioritised over semantic fidelity, or as a preliminary screening tool before targeted ViT-7b analysis of regions of interest. However, for any application requiring reliable semantic segmentation, the ViT-7b is clearly the preferred architecture.

5.2.5 Semantic Search and Zero-Shot Segmentation

The interactive semantic querying tool (Section 4.4) was applied to the 3D feature bank to assess the discriminative quality of the projected embeddings. Figure 16 demonstrates a representative query, where a single pixel selection on a source image propagates through the full 3D model via cosine similarity. The resulting heatmap shows high activation concentrated on geometrically and taxonomically consistent structures, with qualitatively minimal false-positive activation on adjacent substrate classes.



(a) 2D Raw Source Image With Selected Query Point (Red X). (b) Cosine similarity 3DGS heatmap based on similarity to query point feature vector.

Figure 16: Example of resulting cosine similarity heatmap from a selected query point in image (a), demonstrating qualitatively effective segmentation. Points most semantically similar to the query point are coloured brightest yellow. Image (a) is one of the 555 source views of the 3DGS model (b) so is one 2D view of a subsection of the full 3D model.

5.2.6 Inter-Annotator Agreement

To contextualise the results and inform the implementation of the 3D segmentation validation reported in Section 5.2.7, inter-annotator agreement was quantified on the manually labelled dataset described in Section 4.5.3. Fleiss’ Kappa (Eq 29) was calculated globally across all classes and on a per-class basis, with the per-class analysis treated as a one-vs-rest binary agreement score.

Table 4: Inter-annotator agreement: Fleiss’ $\kappa \in [0, 1]$, for overall and per-class agreement. Interpretation thresholds follow Landis and Koch [42]. Per-class values are one-vs-rest binary Fleiss’ κ . N denotes points where at least one annotator assigned that class. Any point labelled unknown is excluded.

Class	Fleiss’ $\kappa \uparrow$	Strength of Agreement	N
Global	0.646	Substantial	406
Coral branching	0.725	Substantial	56
Coral smooth	0.766	Substantial	87
Sand	0.678	Substantial	154
Rock	0.543	Moderate	196
Other living	0.219	Fair	15
Equipment*	1.000	N/A	1

*Due to sample size of $N = 1$, result is unreliable.

Table 4 reports the inter-annotator agreement across both point sets, totalling 600 labelled points. Points labelled unknown by any annotator were removed from analysis, leaving 406 points for this evaluation. Global agreement of $\kappa = 0.646$ indicates substantial agreement among all three annotators. This suggests that the underlying visual classes are sufficiently distinguishable by non-expert annotators in this imagery.

At the per-class level the results are more mixed. The two coral classes and sand were all well agreed upon ($\kappa > 0.67$), suggesting relatively unambiguous visual appearance of these classes in the imagery. Agreement on ‘Rock’ is notably lower ($\kappa = 0.54$), Figure 17 shows that for 52 of the 195 instances where Annotator A labelled a point ‘rock’, Annotator B labelled it ‘sand’. This indicates that the primary reason for this lower agreement is confusion between rock and sand, though there are additional disagreements at the rock–coral boundary. These ambiguities were consistent across all pairings of the three annotators. The consistency of this confusion pattern suggests the ambiguity is primarily intrinsic to the rock-sand and rock-coral boundaries in this imagery, rather than individual misclassification errors.

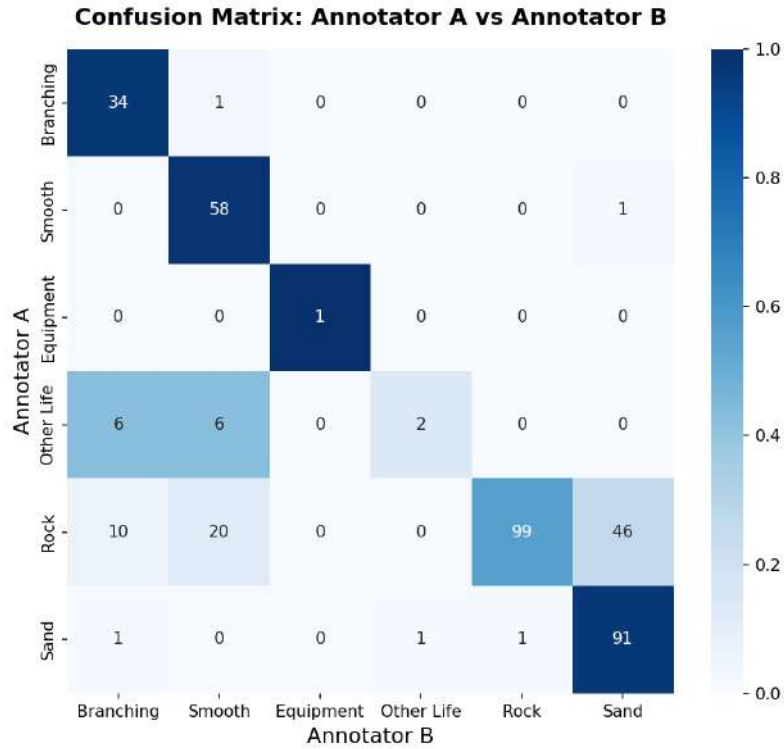


Figure 17: Confusion matrix for a representative annotator pair. Quantities are raw label counts and colours are row-normalised.

The ‘fair’ agreement of the ‘other living’ class ($\kappa = 0.22$) is less reliable due to its small sample size ($N = 15$). At this sample size, κ is highly sensitive to individual decisions. However, the low agreement is likely due to its role as a broad catch-all category. Annotators lacking taxonomic expertise frequently disagreed whether a point fit this label or one of the coral labels. For this reason it was decided to exclude it from the subsequent 3D segmentation validation (Section 5.2.7).

These results demonstrate that the reference labels are broadly reliable for use in downstream segmentation evaluation. Substantial global agreement and high per-class consistency across the coral and sand categories suggest that the labelling protocol was consistent despite lacking trained taxonomists. Two limitations are identified, in particular the rock-sand boundary but also the weak performance of the ‘other living’ class. These limitations are used to contextualise the 3D segmentation performance. Points labelled identically by all three annotators can be treated as high-confidence reference labels throughout the following analysis.

5.2.7 3D Segmentation Validation

Using the reference labels generated following the protocol described in Section 4.5.3, 3D segmentation was evaluated across three-label resolutions (majority vote, full consensus and per individual annotator) and three class granularities (binary, substrate and fine). Table 5 shows that weighted F1 values scaled consistently with the number of reference points per class k , with the largest gains between $k = 1$ and $k = 3$ and reduced benefit subsequently. Variance across random reference-point selections also decreased with k , suggesting that with fewer labels the selection of the points is more significant to create a representative feature prototype.

Table 5: 3D segmentation validation for ViT-7b. Weighted F1 (mean $\pm$ std over 100 seeds) across label-resolution modes, class granularities, and reference counts k . Human ceiling is the mean pairwise percent agreement between annotators at each granularity, computed on the points labelled by all three annotators.

Label set	Granularity	Ceiling	F1 ($k = 1$) $\uparrow$	F1 ($k = 3$) $\uparrow$	F1 ($k = 5$) $\uparrow$	F1 ($k = 10$) $\uparrow$
Majority	Binary	0.936	0.72 ± 0.11	0.79 ± 0.07	0.81 ± 0.05	0.82 ± 0.04
Majority	Substrate	0.933	0.64 ± 0.11	0.73 ± 0.06	0.76 ± 0.04	0.77 ± 0.03
Majority	Fine	0.839	0.55 ± 0.11	0.65 ± 0.06	0.68 ± 0.04	0.70 ± 0.03
Consensus	Binary	1.00	0.77 ± 0.09	0.83 ± 0.05	0.84 ± 0.05	0.85 ± 0.05
Consensus	Substrate	1.00	0.72 ± 0.10	0.79 ± 0.05	0.80 ± 0.04	0.81 ± 0.04
Consensus	Fine	1.00	0.74 ± 0.11	0.82 ± 0.05	0.83 ± 0.03	0.84 ± 0.02
Annotator A	Fine	0.839	0.55 ± 0.10	0.62 ± 0.07	0.64 ± 0.05	0.67 ± 0.03
Annotator B	Fine	0.839	0.57 ± 0.10	0.67 ± 0.06	0.70 ± 0.04	0.72 ± 0.02
Annotator C	Fine	0.839	0.57 ± 0.12	0.69 ± 0.07	0.72 ± 0.04	0.74 ± 0.02

Under the majority vote labels, the model at $k = 10$ reaches 88% of the human ceiling at binary granularity ($0.82/0.936$) and 83% at fine granularity ($0.70/0.839$). The consensus-level human ceiling is trivially 1.00, and as such the model reaches 85% and 84% of the human ceiling for consensus labels. These results suggest that the model consistently performs at 82 – 88% of the expected human performance on the same task.

A consistent gap is observed between the majority-vote and consensus label sets across all granularities and k values. At the fine granularity with $k = 5$, consensus labels yield a weighted F1 of 0.83 compared to 0.68 for majority labels, a difference of 0.15 F1 points. This gap narrows to 0.04 for the binary task (0.84 vs 0.81), where inter-annotator disagreement is lower. Across the results in Table 5, restricting evaluation to the subset of points with consensus reduces label noise and inflates measured performance. This is because the consensus set removes the most ambiguous points, predominantly at the rock–sand and rock–coral boundaries, where annotator disagreement is highest (Section 5.2.6). Therefore, the consensus results should be viewed as an upper bound achievable on unambiguous substrate regions, whereas the majority vote results are the more conservative and practically relevant result, capturing the ambiguous

regions that inevitably exist throughout a real reef. Per-annotator evaluations span 0.67 to 0.74 at $k = 10$, confirming that the majority vote score (0.70) is not driven by any single annotator’s labels.

Table 6, shows the per-class precision, recall and F1 at fine granularity and $k = 5$ majority-vote labels for both architectures. Overall the ViT-7b outperforms the ConvNeXT model-L, consistent with prior 2D analysis. The ViT performs notably better on the rock–sand boundary, likely due to its finer resolution allowing for greater discriminability of the small textural differences between the two less visually distinct classes. However, the two models perform almost identically on the two coral classes, this indicates that on very broad, visually distinct categories ConvNeXT is sufficient and can match the ViT performance. This was supported across the granularities in testing with the ConvNeXT achieving similar performance for the binary and substrate level classifications, but performing notably worse at fine granularity.

Table 6: Per-class classification performance at fine granularity ($k = 5$, majority labels, averaged across 5 seeds) for ViT-7b and ConvNeXT-L.

Class	Precision $\uparrow$		Recall $\uparrow$		F1 $\uparrow$		Support
	ViT	Conv	ViT	Conv	ViT	Conv	
coral_branching	0.65	0.73	0.88	0.79	0.75	0.76	86
coral_smooth	0.76	0.77	0.76	0.73	0.76	0.75	96
rock	0.78	0.63	0.53	0.45	0.63	0.52	200
sand	0.62	0.48	0.78	0.68	0.69	0.57	130
Macro avg	0.70	0.65	0.74	0.66	0.71	0.65	512
Weighted avg	0.71	0.64	0.70	0.62	0.69	0.62	512

The coral classes achieve the highest F1 scores of 0.75 and 0.76, with coral branching having notably high recall (0.88), but reduced precision (0.65), indicating that the feature space creates a generous boundary for the coral branching class. This is consistent with the structural distinctiveness of branching morphologies, their high-frequency surface texture provides a strong discriminative signal relative to flat substrates and smooth corals. However, this complex structure also leads to a greater level of occlusions, leading to features bleeding onto adjacent surfaces and false-positives. This suggests that the occlusion-culling stage is not able to fully resolve fine-scale occlusions in branching structures.

Rock is the weakest performing class ($F1 = 0.63$), with high precision (0.78) but low recall (0.53) indicating that a substantial portion of true rock points were misclassified as other classes. This aligns with the annotator disagreement identified in Section 5.2.6, where the rock–sand boundary accounted for the largest source of inter-annotator disagreement. Rock and sand share similar coarse surface texture in the DINOv3 feature space, and their separation depends on subtle differences that may not be encoded reliably at the 64-dimensional compression level. Sand mirrors this (recall 0.78, precision 0.62), with many predicted sand points corresponding to the true-rock points, further confirming the same sand–rock ambiguity in the opposite direction.

The ‘equipment’ class was excluded from all granularity levels due to insufficient support, only a single labelled point exists across the dataset. The ‘other life’ class was excluded for two reasons. First, the extremely poor agreement of the ‘other life’ class ($\kappa = 0.22$) means the reference labels for the class are inherently unreliable, so subsequent performance metrics would demonstrate label noise more than feature discriminability. Second, the poor agreement further reduces the number of points included after the majority-vote, leaving insufficient support for stable evaluation.

Across label-resolution modes and granularities the model demonstrates 3D segmentation performance that is meaningful but bounded. On the binary substrate-versus coral task the model reaches 88% of the human ceiling at $k = 10$, and at fine granularity it reaches 83%, with remaining error concentrated on the rock–sand boundary which drives the majority of inter-annotator disagreement. These results support the fact that the model is able to reliably discriminate between distinct visual morphologies within a reef with very few reference label points. Furthermore, it confirms that the 3D projection pipeline preserves the discriminative ability of the underlying 2D features.

5.2.8 Pipeline Ablation

To quantify the contribution of the projection pipeline’s engineering components, a naive baseline was compared against the full pipeline following the protocol described in Section 4.5.5. Both conditions were evaluated using identical source features. The results are summarised in Table 7.

The full pipeline achieves marginally higher split-half consistency across all metrics (mean: 0.887 vs 0.897, median: 0.910 vs 0.924), and modestly higher silhouette scores at most values of k ($\Delta = 0.013$ – 0.024). While these improvements are consistent in direction, their magnitude is small, suggesting that the DINOv3 embeddings are inherently robust to projection noise. Even without occlusion culling, confidence weighting, or guided filter upsampling, multi-view averaging over dense survey coverage produces a reliable semantic representation. The most notable improvement is the percent above 0.9 ($\Delta = 7.2\%$), suggesting the primary benefit of the pipeline is stabilising the lower tail of the consistency distribution rather than shifting it uniformly.

Table 7: Ablation comparing naive and full projection pipelines. The naive baseline uses bilinear upsampling, no occlusion culling, uniform averaging, no feature centring, and no reliability filtering. Split-half results are aggregated over five random splits. Few-shot F1 (weighted) is reported under majority-vote label resolution at the fine granularity level. $\uparrow$ indicates higher is better.

Metric	Naive	Full	Δ
Feature Stability			
Split-half mean $\uparrow$	0.887	0.897	+0.010
Split-half median $\uparrow$	0.910	0.924	+0.014
Percent > 0.9 $\uparrow$	55.6%	62.8%	+7.2%
Cluster Separability			
Silhouette ($k=2$) $\uparrow$	0.279	0.292	+0.013
Silhouette ($k=4$) $\uparrow$	0.285	0.299	+0.014
Silhouette ($k=8$) $\uparrow$	0.256	0.279	+0.024
Few-shot Classification (fine)			
F1 weighted ($k=1$) $\uparrow$	0.53	0.55	+0.02
F1 weighted ($k=5$) $\uparrow$	0.66	0.68	+0.02
F1 weighted ($k=10$) $\uparrow$	0.68	0.70	+0.02

These gains translate to a consistent +0.02 improvement in few-shot weighted F1 at the fine granularity level, stable across all k values (Table 5). This gain is uniform across k , granularity and annotator label sets, indicating a general improvement in feature quality rather than an artefact of a particular evaluation.

The modest magnitude of all improvements suggests feature reliability is primarily driven by the density of the survey coverage rather than the sophistication of the aggregation strategy. Each Gaussian is observed by a median of 68 views, so with many views both weighted and unweighted averaging converge to a stable estimate. Given that the additional computational cost of the full pipeline is modest relative to the upstream photogrammetry reconstruction and feature extraction stages, the engineering components are retained as a low-cost refinement that provides consistent modest improvements. Though these results suggest that for well-surveyed reef sites the quality of the underlying features is the key factor over the sophistication of the projection.

6 Discussion

The results presented in this report support the central hypothesis that DINOv3 embeddings can be projected into an explicit 3DGS representation to produce a semantically discriminative and reliable digital twin of a coral reef without labelled training data. The 3D segmentation validation confirms that the semantic features are ecologically valid, achieving up to 83% of the human performance ceiling for fine class segmentation at $k = 10$.

This directly addresses a key gap identified in the literature. While previous work has demonstrated strong performance of foundation model embeddings in 2D coral analysis, and separate advances in 3D reconstruction have enabled detailed geometric modelling, to our knowledge, no existing approach has successfully integrated dense, high-dimensional semantic features into an explicit 3D representation suitable for label-efficient reef analysis. The results presented here provide evidence that this approach is feasible, although validation is currently limited to a single reef site.

This section contextualises these findings, identifies limitations of the current work and suggests directions for future development.

6.1 Architectural Comparison: ViT-7b versus ConvNeXT-L

The comparative results across both 2D and 3D stages reveal a consistently stronger performance from the ViT-7b architecture. In 2D, this is reflected by sharper boundary preservation and finer semantic resolution, attributable to the ViT’s smaller patch stride (16 versus 32) and higher embedding dimensionality (4096 vs 1536). Once projected into 3D, the performance gap widens further due to a fundamental difference in how the two architectures respond to the systematic illumination gradients characteristic of underwater survey imagery.

The ViT’s global self-attention mechanism conditions each patch token on the full image context, providing implicit normalisation against local photometric variation. In contrast, the ConvNeXT’s locally bounded receptive field encodes these gradients directly into the feature representation, producing the spatial banding artifacts observed in the 3D semantic model. Naseer et al. [43] demonstrate that ViTs exhibit greater robustness against ‘common natural corruptions’ such as noise blur and contrast and pixelation artefacts, which aligns with the illumination resilience observed in the ViT output here. The underwater setting amplifies this architectural distinction because the water column introduces spatially correlated noise that is precisely what ViTs are shown to be more robust to.

The 3D segmentation results further support this, with performance notably lower when separating the more visually ambiguous coral–sand boundary for the ConvNeXT, where subtle textural differences must be distinguished while competing against systematic illumination variation. However, performance on the two coarse coral classes was highly similar between the architectures, indicating that the ConvNeXT is sufficient when class boundaries are strongly visually distinct.

The ConvNeXT’s approximately $60\times$ faster processing time also remains a significant practical advantage. From this, the ConvNeXT appears preferable when class boundaries are distinct, such as binary coral versus non-coral classification as it achieves comparable performance with greatly reduced computational expense.

One direction that may improve the ConvNeXT’s 3D performance is illumination normalisation as a pre-processing step. Underwater colour correction methods such as the Sea-thru algorithm [44] explicitly remove the depth dependent colour cast and attenuation introduced by the water column. Applying such a correction prior to feature extraction would reduce the photometric covariance that the ConvNeXT encodes into its features, potentially recovering much of the view-invariance that the ViT achieves through architectural means. If this gap could be closed, the ConvNeXT architecture would offer a compelling lightweight alternative for large scale rapid surveys where the ViT’s computational cost is prohibitive.

6.2 Underwater Domain Generalisation and Comparison with Existing Approaches

The strong 2D and 3D performance of the DINOv3 in the underwater domain is notable given that the model was trained entirely on terrestrial imagery and has received no domain-specific fine-tuning. This suggests that the rich hierarchical representations learned from large-scale self-supervised pretraining generalise effectively across the domain gap introduced by the underwater setting, including water column attenuation and backscatter. The ability of a general purpose model to bridge this gap without adaptation is practically significant as it removes the dependency on large domain-specific labelled datasets that has historically constrained supervised approaches to coral monitoring [21].

This finding extends the observations of Raine et al. [11], who demonstrated that DINOv2 embeddings are effective for coral segmentation in sparse-label settings. This work advances this by evaluating the more recent DINOv3 architecture, which has not been applied to coral imagery previously. Moreover, it demonstrates that these features remain discriminative when projected into a 3D model representation. This is practically beneficial as 3D reconstructions can capture important metrics (such as colony volume) for reef assessment [13] that can then be calculated on a coral segmented basis which would not be possible using 2D methods. The few-shot classification results (Table 5) confirm that meaningful segmentation performance is achieved from only a single reference point per class, exceeding the label sparsity and performance of Raine et al. [11]. Notably this is achieved with far fewer classes and point based classification as opposed to mask based, meaning stronger performance is expected. Nonetheless, it confirms that DINOv3 features projected into 3D can discriminate between reef morphologies under highly limited labels.

The 2D few-shot results (Table 1) demonstrate the raw DINOv3 features achieve mIoU comparable to the denoised DINOv2 baseline of Raine et al. [11] across all tested label counts. Although, it is important to note methodological differences that limit direct comparison. Their results are evaluated on the UCSD Mosaics dataset across 33 classes using denoised DINOv2 features, whereas our evaluation uses 13 classes on the Moorea dataset with raw DINOv3 features. Having fewer classes would be expected to inflate the mIoU, while the absence of denoising would be expected to reduce it. Furthermore, their labels are propagated into a full mask based classification task, as opposed to a point based one, which increases the volume of visually ambiguous boundary cases, reducing their mIoU scores. With this in mind, the performance should solely be interpreted as a demonstration that DINOv3 is capable of comparable few-shot segmentation to existing baselines, not that it is explicitly better. It also suggests that this performance could be significantly improved with smart labelling strategies as demonstrated by Raine et al. [11] (which substantially boost their reported mIoU scores), or denoising of the DINOv3 features.

This performance is also notable when compared to CoralSCOP [24] which achieves strong dense segmentation, but relies on a massive dataset of 1.3 million masks and operates in 2D. Similarly, Sauder et al. [30] demonstrate successful 2D-to-3D projection of coral segmentation maps, but their approach projects discrete class labels from a fully trained task-specific segmentation model. In contrast, this pipeline demonstrates that the high-dimensional feature space of DINOv3 is sufficiently robust to achieve segmentation through simple cosine similarity in 3D space, without the need for domain specific fine tuning. Furthermore, this approach provides much greater analytical flexibility, supporting arbitrary downstream queries as demonstrated in Section 5.2.5. In contrast, discrete labels would need to be retrained and re-projected to identify a novel class definition.

The GauSSmart system [31] uses DINOv3 features as a training-time loss but does not store them explicitly, preventing semantic querying. Feature-3DGS [32] non-linearly reduces 2D features into a compressed student representation optimised for rendering speed, but this sacrifices a large amount of the discriminative depth of the original embedding space. Both methods, however, support novel-view rendering of semantic fields, which the projection approach used here does not. Because features are attached to existing Gaussian primitives, rather than learned as a function of viewing direction, the semantic field can only be queried at positions of existing Gaussians, not arbitrary novel viewpoints. Where the goal is fixed, accurate segmentation, this static approach is acceptable, if real-time rendering was needed approaches more similar to theirs would likely be preferred. Neither GauSSmart nor Feature-3DGS has been applied to the underwater domain, and they do not prioritise discriminative depth that is needed for coral segmentation. Although our approach does apply PCA compression, this was selected to retain the majority of the semantic variance and preserve the linear structure of the feature space to enable cosine-similarity based querying and few-shot classification demonstrated in the results. This pipeline could be used without PCA compression to truly maximise the discriminative depth of the feature space, however this was not computationally feasible in the scope of this work.

Despite the promising results, the sensitivity of the ConvNeXT architecture to illumination gradients suggests that underwater specific properties are not fully compensated for by general pretraining alone. This and further performance gains for classification of more specific morphologies motivates the development of a vision foundation model trained specifically on underwater or coral imagery. Such a model could better account for the spectral properties of the water column, the structural regularities of coral morphologies, and the characteristic lighting variability of photogrammetric surveys.

There are multiple approaches to training a domain-specific vision foundation model that would likely

enhance performance. One possibility is fine-tuning DINOv3 on a large corpus of unlabelled underwater imagery using the existing self-supervised objective which would adapt the feature space to the target domain without requiring labels. Second, training a dedicated autoencoder on the backbone of DINOv3 could provide a more compressed, domain optimised representation suited to the specific demands of coral analysis. Either approach would likely improve both the view-invariance of extracted features and the discriminability of the fine-grained taxonomic distinctions that the general model currently struggles to separate. A domain-specific ViT therefore represents a natural and high-impact direction for future development of this framework.

6.3 Projection Pipeline Reliability

The high split-half consistency results (Section 5.2.3) confirm that the projection pipeline produces stable feature representations when given equivalent subsets of input views. This result provides evidence that the aggregation strategy is robust to the specific subset of source images used, which has practical significance for repeat survey scenarios where image coverage may vary between visits. The high consistency also implies that the full image set may not be necessary for adequate performance. To determine this, a systematic study varying the number of input images and measuring the resulting feature stability would quantify the minimum survey density required to inform more efficient survey protocols.

However, the split-half test measures whether the pipeline produces consistent output given matched input conditions, it does not verify that the projected features faithfully preserve the semantic quality of the underlying 2D embeddings since both halves may produce the same systematic errors. The primary evidence for semantic validity once projected into 3D is the segmentation validation test (Section 5.2.7).

The ablation study of the pipeline demonstrates that the processing steps outlined (Section 4.3.3 - 4.3.6) provide a small, but consistent improvement to both the discriminability and reliability of the resulting features. However, the Tabuhan dataset represents a rigorous photogrammetric survey, where every Gaussian in the sub-block is observed by a large number of source images. Under these conditions simple averaging already converges to a stable estimate, and the confidence weighting, occlusion culling and guided filter upsampling provide diminishing returns. The more sophisticated pipeline is expected to contribute more substantially under sparser survey conditions, where each Gaussian may be observed by only a handful of views and occlusion is more likely to introduce systematic bias. Testing under progressively reduced survey density would verify this hypothesis and provide practical guidance on the minimum survey coverage required for reliable semantic projection.

Given that the additional computational cost of the full pipeline is modest relative to the upstream ViT feature extraction stage (which dominates the total processing time) retaining the full pipeline as a low-cost refinement is justified. The consistent direction of improvement across nearly all evaluated conditions, suggests that the components collectively improve the discriminability of the feature field even when the magnitude is small. Furthermore, the upsampling strategy could be improved in future work through the application of learned feature upsampling methods such as FeatUp [45], which has demonstrated strong performance for DINOv2 features and would likely transfer to DINOv3, particularly at morphological boundaries where guided filtering currently produces the most residual blur.

The 3D results stem from a feature bank compressed to 64 dimensions via PCA, retaining 84.4% of the explained variance of the ViT-7b feature space. Whether this compression level represents an optimal trade-off between semantic fidelity and storage efficiency remains an open question. The Kneedle algorithm identified an elbow at 29 principal components, suggesting that the primary structural variance is captured well below the chosen cut-off. However, the conservative choice of 64 dimensions was motivated by preserving the long-tail of fine-grained local features that may be critical for taxonomic discrimination. Systematically varying the compression level and measuring the resulting silhouette scores and few-shot classification performance would empirically determine the minimum dimensionality sufficient for the intended downstream tasks, potentially enabling further storage reductions without meaningful loss of semantic quality.

6.4 Manually Labelled 3D Validation Dataset

The inter-annotator agreement analysis (Section 5.2.6) confirms that the four classes ultimately used for 3D segmentation evaluation (coral branching/smooth, sand and rock) were consistently categorised by all three annotators, with global Fleiss’ $\kappa = 0.646$ and per-class $\kappa > 0.54$ for all retained classes. These fall in the substantial and moderate agreement categories respectively that are proposed by Landis and Koch [42]. They are widely used though not derived from any formal statistical analysis; nonetheless, they provide a simple framework to contextualise agreement levels. The consistency of the rock–sand confusion pattern across all annotator pairings (e.g. Figure 17) suggests that remaining disagreement is largely a property of the imagery rather than primarily labelling error. Overall, this provides confidence that the retained labels represent a reliable, but coarse, ground truth for evaluation.

Using these labels, the 3D segmentation results (Table 5) provide quantitative confirmation that the projection pipeline preserves discriminative power. The model reaches 88% of the human performance ceiling at binary granularity and 83% at fine granularity, with only ten reference points per class. These metrics establish that the 3D feature bank reliably separates the broad morphological categories present in this reef sub-block.

However, several limitations constrain the scope of this validation. None of the three annotators are trained taxonomists, which necessitated the deliberately coarse class taxonomy used. Species-level distinctions, which would be required for detailed ecological assessment, could not be applied consistently so were not performed.

The validation was limited to 600 labelled points, with further visually ambiguous points ultimately excluded due to being labelled ‘unknown’. This constrained the reliability of analysis, since following the inter-annotator agreement testing only four classes remained (coral branching/smooth, sand and rock). The ‘other living’ class was excluded due to poor inter-annotator agreement ($\kappa = 0.22$) and the ‘equipment’ class due to only having a single labelled point. With a greater number of labels, both of these classes may have presented more reliable results and been included in analysis. Therefore, the resulting four-class evaluation tests the model’s ability to discriminate broad rather than fine taxonomic categories.

Furthermore, the annotators include the author and their two supervisors. Although measures were taken to avoid bias such as performing labelling without access to one-another’s labels, some bias is likely unavoidable as annotators had previously viewed PCA outputs of the feature space (e.g. Figure 13). Therefore, implicit bias towards qualitative clusters in the feature space cannot be ruled out.

These limitations motivate a more rigorous validation conducted using domain experts and species-level classifications over a larger quantity of points as a clear direction for future work. This would allow for testing of the model’s ability to categorise more refined visual properties, a reduction of bias and more consistent labelling across annotators.

6.5 Ecological Utility and Practical Application

The unsupervised clustering results provide qualitative evidence that the projected feature field encodes ecologically meaningful structure. At $k = 6$, the ViT-7b cluster assignments align closely with visually identifiable substrate boundaries in the RGB reference, with distinct clusters corresponding to recognisable coral morphologies and sand substrate. The visual coherence of the clusters suggests that the semantic 3DGS representation captures at minimum the coarse ecological partitions relevant to reef monitoring. The binary clustering result at $k = 2$ is particularly encouraging, the ViT-7b produces a clean coral versus substrate partition that covers the visible reef extent accurately. This, along with the strong binary segmentation results, suggests immediate utility for automated coral coverage estimation which is a fundamental metric in reef health assessment [12]. Combined with the volumetric capabilities of the 3DGS representation, this would enable quantitative tracking of total coral volume across repeat surveys without any manual annotation, providing a practical baseline monitoring metric for restoration projects. This aligns with the direction advocated by Ferrari et al. [15], who identify photogrammetric monitoring as essential to capturing the structural complexity metrics that 2D surveys cannot provide.

The semantic search tool provides a strong proof of concept for interactive human-in-the-loop reef analysis.

The ability to propagate a single pixel query across the full 3D feature bank via cosine similarity enables rapid identification of semantically similar structures throughout the reef without any additional labelling or model training. Importantly, once the initial feature extraction and projection pipeline has been run, the semantic querying and segmentation is computationally inexpensive, similarity scores across the full Gaussian feature bank can be computed in seconds on standard hardware using vectorised cosine similarity. This lowers the barrier to entry for reef monitoring practitioners who may not have access to powerful computational resources, addressing the accessibility concerns raised by Burns, Bollard, and Narayanan [21] in relation to the deployment of AI tools in ecological settings. The computationally intensive stages of ViT feature extraction and 3D projection only need to be run once per survey meaning the interactive query interface operates entirely on precomputed embeddings.

These tools have several practical applications beyond simple visualisation. First, it would substantially improve the efficiency of manual annotation workflows by highlighting candidate regions for expert review, reducing the search space from the full model to a semantically filtered subset. Second, a taxonomist could iteratively refine the semantic boundaries of clusters by confirming or rejecting highlighted regions, providing a lightweight human-in-the-loop mechanism for improving segmentation quality without exhaustive point labelling.

More speculatively, the approach is not inherently limited to taxonomic segmentation. The DINOv3 feature space may encode additional ecologically relevant properties beyond morphological identity. For example, coral bleaching produces a spectral shift from pigmented to pale tissue that represents a visually distinct appearance [46]. Whether this shift is preserved within the retained 64 principal components would need to be identified and is further cause for investigating this approach across varying compression levels. Providing a small number of example features from known bleached coral structures as query vectors could enable systematic identification of bleaching across the full 3D model, representing a compelling application for reef health monitoring that warrants dedicated investigation.

These applications are currently demonstrated on a single 5×5 m sub-block of one reef site. While this sub-block was selected to capture a representative diversity of morphologies, the generalisability of the pipeline to environments with varying structural complexity, species composition, or survey methodology remains unvalidated. Scaling to full reef sites (hundreds to thousands of square metres) will require optimisation of the projection pipeline.

6.6 Future Work

The limitations identified throughout this discussion suggest several prioritised directions for future development of this work. These would all ideally be informed by domain expert collaboration to ensure practical domain specific considerations are incorporated.

The highest priority is rigorous 3D validation using species-level annotation from trained marine taxonomists, which would directly establish whether the 3D feature space captures fine taxonomic distinctions and provide a high-quality benchmark for validating any further work.

Multi-site evaluation across geographically diverse reef systems and repeated temporal surveys is essential to establish the spatio-temporal generalisability that Burns, Bollard, and Narayanan [21] outline as a critical target for scalable monitoring. The split-half consistency results provide preliminary evidence that the feature space tolerates variation in input imagery, but dedicated temporal validation would be required.

Domain adaptation of the DINOv3 backbone would likely be the most substantial boost to performance of this methodology (Section 6.2). However, learned feature upsampling methods such as FeatUp [45] also represent promising improvements to this pipeline. Finally, integrating the semantic query tools into an accessible, web-based interface would increase accessibility for marine ecologists who may lack computational expertise.

7 Conclusion

This report has presented a novel pipeline for fusing dense semantic embeddings from the DINOv3 self-supervised foundation model with 3D Gaussian Splatting to produce a semantically queryable digital twin of a coral reef without any labelled training data. The work aims to address the critical gap between the abundance of high-quality photogrammetric data now available for coral reef monitoring and the lack of scalable, label-efficient tools for extracting ecologically meaningful information from it.

Two-dimensional validation on the Moorea Labelled Corals dataset confirms that DINOv3 embeddings capture semantically meaningful structure across species level distinctions in underwater coral imagery without domain specific fine-tuning ($F1 = 0.58$). The ViT-7b consistently outperforms the ConvNeXT-L across all evaluated metrics, with the performance gap attributable to the ViT’s global self-attention mechanism providing resilience to systematic noise within underwater survey imagery.

Projecting these features into 3D via the established pipeline yields a semantically coherent feature field across 2.8 million Gaussian primitives, with mean split-half similarity of 0.899 confirming stability with varying and limited input data. Few-shot segmentation against manually labelled reference points reaches 83% of the human performance ceiling for the fine class granularity (smooth coral, branching coral, sand and rock) using only ten reference points per class. This provides quantitative evidence that the projection preserves the discriminative power of the underlying 2D features. Unsupervised clustering produces ecologically interpretable partitions, and a single-query semantic search tool demonstrates qualitatively promising zero-shot 3D segmentation from a single pixel selection.

The primary limitations are the restriction to a single reef site and the use of a coarse, non-expert labelled classes for 3D validation. Addressing these through expert taxonomic annotation and multi-site evaluation would be the most important areas for future work. Beyond this, domain adaptation of the DINOv3 backbone and temporal change detection across repeat surveys represent promising directions for extending the framework towards a viable monitoring tool for practical and scalable deployment.

Overall, these results establish DINOv3 and 3DGS as a viable foundation for label-efficient coral reef monitoring, with a clear path from the current proof of concept to a tool that could meaningfully support reef conservation and restoration at scale.

References

- [1] Ove Hoegh-Guldberg et al. “Coral reefs under rapid climate change and ocean acidification”. In: *science* 318.5857 (2007), pp. 1737–1742.
- [2] Don E. McAllister et al. “Mapping and GIS analysis of the global distribution of coral reef fishes on an equal-area grid”. In: *Mapping the Diversity of Nature*. Dordrecht: Springer Netherlands, 1994, pp. 155–175. ISBN: 978-94-011-0719-8. DOI: 10.1007/978-94-011-0719-8_10. URL: https://doi.org/10.1007/978-94-011-0719-8_10.
- [3] Fredrik Moberg and Carl Folke. “Ecological goods and services of coral reef ecosystems”. In: *Ecological Economics* 29.2 (1999), pp. 215–233. ISSN: 0921-8009. DOI: [https://doi.org/10.1016/S0921-8009\(99\)00009-9](https://doi.org/10.1016/S0921-8009(99)00009-9). URL: <https://www.sciencedirect.com/science/article/pii/S0921800999000099>.
- [4] Lijing Cheng et al. “Record high temperatures in the ocean in 2024”. In: *Advances in Atmospheric Sciences* 42.6 (2025), pp. 1092–1109.
- [5] Clive R Wilkinson. “Global and local threats to coral reef functioning and existence: review and predictions”. In: *Marine and freshwater research* 50.8 (1999), pp. 867–878.
- [6] Yves-Marie Bozec et al. “A rapidly closing window for coral persistence under global warming”. In: *Nature Communications* 16.1 (2025), p. 9704.
- [7] Bernhard Kerbl et al. “3d gaussian splatting for real-time radiance field rendering.” In: *ACM Trans. Graph.* 42.4 (2023), pp. 139–1.
- [8] Oriane Siméoni et al. “Dinov3”. In: *arXiv preprint arXiv:2508.10104* (2025).

- [9] Mathilde Caron et al. “Emerging properties in self-supervised vision transformers”. In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2021, pp. 9650–9660.
- [10] Maxime Oquab et al. *DINOv2: Learning Robust Visual Features without Supervision*. 2024. arXiv: 2304.07193 [cs.CV]. URL: <https://arxiv.org/abs/2304.07193>.
- [11] Scarlett Raine et al. “Human-in-the-loop segmentation of multi-species coral imagery”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.
- [12] Jos Hill and Clive Wilkinson. “Methods for ecological monitoring of coral reefs”. In: *Australian Institute of Marine Science, Townsville* 117 (2004).
- [13] Renata Ferrari et al. “3D photogrammetry quantifies growth and external erosion of individual coral colonies and skeletons”. In: *Scientific reports* 7.1 (2017), p. 16737.
- [14] Jonathan Teague et al. “A Review of Current and New Optical Techniques for Coral Monitoring”. English. In: *Oceans* 3.1 (Jan. 2022), pp. 30–45. ISSN: 2673-1924. DOI: 10.3390/oceans3010003.
- [15] Renata Ferrari et al. “Photogrammetry as a tool to improve ecosystem restoration”. In: *Trends in Ecology & Evolution* 36.12 (2021), pp. 1093–1101.
- [16] Matan Yuval et al. “Repeatable Semantic Reef-Mapping through Photogrammetry and Label-Augmentation”. In: *Remote Sensing* 13.4 (2021). ISSN: 2072-4292. DOI: 10.3390/rs13040659. URL: <https://www.mdpi.com/2072-4292/13/4/659>.
- [17] R. Jannathul Firdous and S. Sabena. “A novel approach to coral species classification using deep learning and unsupervised feature extraction”. In: *Journal of Spatial Science* 0.0 (2024), pp. 1–28. DOI: 10.1080/14498596.2024.2383881. eprint: <https://doi.org/10.1080/14498596.2024.2383881>. URL: <https://doi.org/10.1080/14498596.2024.2383881>.
- [18] Oscar Beijbom et al. “Automated annotation of coral reef survey images”. In: *IEEE Conference on Computer Vision and Pattern Recognition* (2012). DOI: 10.1109/CVPR.2012.6247798.
- [19] Andrew King, Suchendra M Bhandarkar, and Brian M Hopkinson. “A comparison of deep learning methods for semantic segmentation of coral reef survey images”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*. 2018, pp. 1394–1402.
- [20] Ming Li et al. “A survey on underwater coral image segmentation based on deep learning”. In: *Geo-spatial Information Science* 28.2 (2025), pp. 472–496. DOI: 10.1080/10095020.2024.2343323. eprint: <https://doi.org/10.1080/10095020.2024.2343323>. URL: <https://doi.org/10.1080/10095020.2024.2343323>.
- [21] Christopher Burns, Barbara Bollard, and Ajit Narayanan. “Machine-learning for mapping and monitoring shallow coral reef habitats”. In: *Remote Sensing* 14.11 (2022), p. 2666.
- [22] Alexander Kirillov et al. “Segment anything”. In: *Proceedings of the IEEE/CVF international conference on computer vision*. 2023, pp. 4015–4026.
- [23] Amy Stone et al. “Mapping emergent coral reefs: a comparison of pixel-and object-based methods”. In: *Remote Sensing in Ecology and Conservation* 11.1 (2025), pp. 20–39.
- [24] Ziqiang Zheng et al. “Coralscop: Segment any coral image on this planet”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024.
- [25] Johannes L. Schonberger and Jan-Michael Frahm. “Structure-From-Motion Revisited”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2016.
- [26] Ben Mildenhall et al. “Nerf: Representing scenes as neural radiance fields for view synthesis”. In: *Communications of the ACM* 65.1 (2021), pp. 99–106.
- [27] Shuai Liu et al. “A Survey of 3D Reconstruction: The Evolution from Multi-View Geometry to NeRF and 3DGS”. In: *Sensors* 25.18 (2025). ISSN: 1424-8220. DOI: 10.3390/s25185748. URL: <https://www.mdpi.com/1424-8220/25/18/5748>.
- [28] Sharath Girish, Kamal Gupta, and Abhinav Shrivastava. *EAGLES: Efficient Accelerated 3D Gaussians with Lightweight EncodingS*. 2024. arXiv: 2312.04564 [cs.CV]. URL: <https://arxiv.org/abs/2312.04564>.
- [29] Aoran Xiao et al. *A Survey of Label-Efficient Deep Learning for 3D Point Clouds*. 2024. arXiv: 2305.19812 [cs.CV]. URL: <https://arxiv.org/abs/2305.19812>.
- [30] Jonathan Sauder et al. *Scalable Semantic 3D Mapping of Coral Reefs with Deep Learning*. 2023. arXiv: 2309.12804 [cs.CV]. URL: <https://arxiv.org/abs/2309.12804>.

- [31] Alexander Valverde et al. “GauSSmart: Enhanced 3D Reconstruction through 2D Foundation Models and Geometric Filtering”. In: *arXiv preprint arXiv:2510.14270* (2025).
- [32] Shijie Zhou et al. “Feature 3dgs: Supercharging 3d gaussian splatting to enable distilled feature fields”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024, pp. 21676–21685.
- [33] Moorea Coral Reef LTER and P. Edmunds. *MCR LTER: Coral Reef: Computer Vision: Moorea Labeled Corals ver 3*. 2019. DOI: 10.6073/pasta/88dde0e68ab5232a470389f4bedd1892. URL: <https://doi.org/10.6073/pasta/88dde0e68ab5232a470389f4bedd1892>.
- [34] Sergei Nozdrenkov et al. *sweet-corals (Revision bd316dc)*. 2025. DOI: 10.57967/hf/5162. URL: <https://huggingface.co/datasets/wildflow/sweet-corals>.
- [35] Nathan Godey, Éric Clergerie, and Benoit Sagot. “Anisotropy is inherent to self-attention in transformers”. In: *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. 2024, pp. 35–48.
- [36] Richard Hartley and Andrew Zisserman. *Multiple view geometry in computer vision*. Cambridge university press, 2003.
- [37] Paulius Micikevicius et al. *Mixed Precision Training*. 2018. arXiv: 1710.03740 [cs.AI]. URL: <https://arxiv.org/abs/1710.03740>.
- [38] Kaiming He, Jian Sun, and Xiaoou Tang. “Guided image filtering”. In: *IEEE transactions on pattern analysis and machine intelligence* 35.6 (2012), pp. 1397–1409.
- [39] Chris Buehler et al. “Unstructured lumigraph rendering”. In: *Proceedings of the 28th annual conference on Computer graphics and interactive techniques*. 2001, pp. 425–432.
- [40] Peter J. Rousseeuw. “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis”. In: *Journal of Computational and Applied Mathematics* 20 (1987), pp. 53–65. ISSN: 0377-0427. DOI: [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7). URL: <https://www.sciencedirect.com/science/article/pii/0377042787901257>.
- [41] Joseph L Fleiss. “Measuring nominal scale agreement among many raters.” In: *Psychological bulletin* 76.5 (1971), p. 378.
- [42] J Richard Landis and Gary G Koch. “The measurement of observer agreement for categorical data”. In: *biometrics* (1977), pp. 159–174.
- [43] Muhammad Muzammal Naseer et al. “Intriguing properties of vision transformers”. In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 23296–23308.
- [44] Derya Akkaynak and Tali Treibitz. “Sea-Thru: A Method for Removing Water From Underwater Images”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2019.
- [45] Stephanie Fu et al. “Featup: A model-agnostic framework for features at any resolution”. In: *arXiv preprint arXiv:2403.10516* (2024).
- [46] G Winters et al. “Photographic assessment of coral chlorophyll contents: Implications for ecophysiological studies and coral monitoring”. In: *Journal of Experimental Marine Biology and Ecology* 380.1-2 (2009), pp. 25–35.